

3

MEASUREMENT AND SPACE

SURFACE AREA AND VOLUME

Some theme parks have wave pools, which are big swimming pools that simulate the movement of the water at a beach. A large volume of water is quickly released into one end of the pool, which produces a large wave that moves from one end of the pool to the other. The excess water in the pool is recycled so that it can be used to produce more waves.



Chapter outline

		Working mathematically				
3.01	Absolute error and percentage error	U	F		R	C
3.02	Rounding error	U	F		R	C
3.03	Areas of composite shapes	U	F	PS	R	
3.04	Surface area of a prism	U	F	PS	R	C
3.05	Surface area of a cylinder	U	F	PS	R	C
3.06	Surface area of a pyramid*	U	F	PS	R	
3.07	Surface areas of cones and spheres*	U	F	PS	R	C
3.08	Surface areas of composite solids*	U	F	PS	R	
3.09	Volumes of prisms and cylinders	U	F	PS	R	
3.10	Volumes of pyramids, cones and spheres*	U	F	PS	R	C
3.11	Volumes of composite solids*	U	F	PS	R	

*PATH

Wordbank

absolute error The greatest possible error of a measurement, equal to half of the precision (smallest unit) on the measuring instrument's scale

cross-section A 'slice' of a solid, taken across the solid rather than along it

curved surface area The area of the curved surface of a solid such as a cylinder or sphere. For example, the curved surface of a cylinder is a rectangle when flattened.

pyramid A solid with a polygon for a base and triangular faces that meet at a point called the apex

rounding error The difference between a calculated answer and an exact answer due to rounding in measurement and calculations

sector A region of a circle cut off by 2 radii, shaped like a piece of pizza

slant height The height of a pyramid or a cone from its apex to its base along a side face rather than its perpendicular height

surface area The total area of all faces of a solid shape

Quiz
Wordbank 3

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (10):

3.04 Surface area of a prism
• Surface areas of prisms

3.05 Surface area of a cylinder

3.06 Surface area of a pyramid

3.07 Surface area of a cone and sphere
• Surface area and volume of a cone

3.09 Capacity of a cylinder
• Volumes of prisms and cylinders

3.10 Volume of a pyramid
• Surface area and volume of a cone
• Volume of a sphere

Twig videos (6):

3.01 Decimal places: Photo finish

3.08 The power of the Sun

3.09 Bees and their hives • The incredible strength of ants • The Menger sponge

3.10 The Pacific Flyer

Quizzes (5):

- Wordbank 3
- SkillCheck 3
- Mental skills 3
- Language of maths 3
- Test yourself 3

Skillsheets (2):

SkillCheck Solid shapes • What is volume?

Worksheets (10):

3.01 Accuracy in measurement 2

3.03 A page of circular shapes

3.04 Surface area 2 • Nets of solids

3.08 A page of prisms and cylinders

3.09 A page of prisms and cylinders • Volumes of solids • Back-to-front problems • Volume and capacity • Biggest volume

3.10 Back-to-front problems (Advanced)

Mind map Surface area and volume

Puzzles (4):

3.03 Area

3.05 Surface area

3.09 Formula matching game

Language of maths Surface area and volume crossword

Technology (2):

3.09 Volume of a box
• Approximating the volume of a cone

Spreadsheet (1):

3.09 Volume of a box

Presentation (1):

3.11 Volumes of shapes

Nelson MindTap

To access resources above, visit
cengage.com.au/nelsonmindtap

In this chapter you will:

- calculate the absolute error and percentage error of a measurement
- identify and investigate rounding error
- calculate the areas of triangles, quadrilaterals, circles, sectors and composite shapes
- calculate the surface areas, volumes and capacities of right prisms and cylinders
- (PATH) calculate the surface areas, volumes and capacities of pyramids, cones, spheres and composite solids

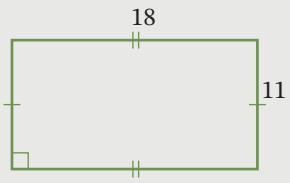
SkillCheck ANSWERS ON P. 583

Quiz
SkillCheck 3

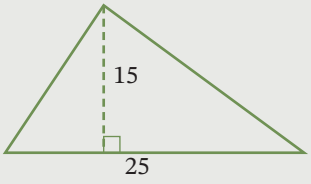
3

1 Calculate the area of each shape. All measurements are in centimetres.

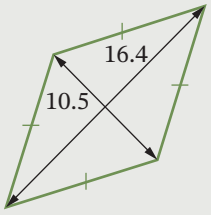
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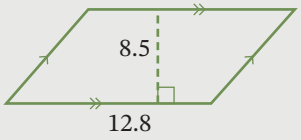
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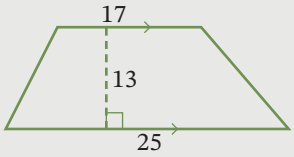
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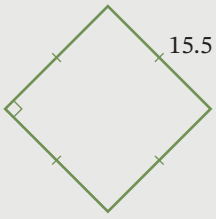
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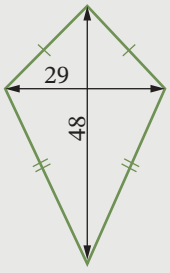
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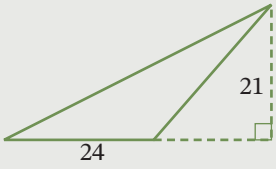
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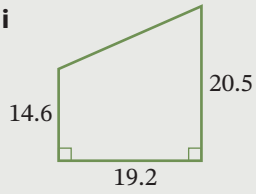
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h

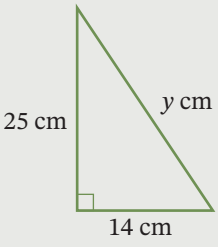


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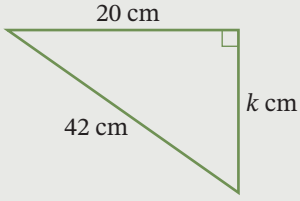


2 Use Pythagoras' theorem to find, correct to one decimal place, the value of each variable.

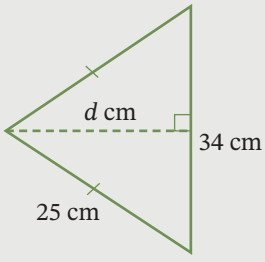
a



b



c



3 For each circle, find correct to one decimal place:

i its circumference **ii** its area.

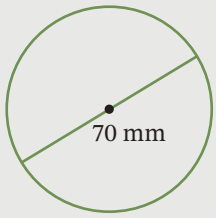
a



b

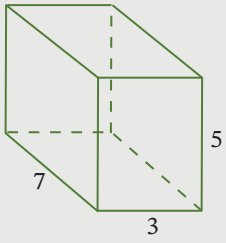


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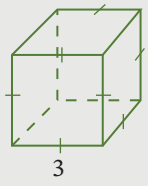


4 Calculate the volume of each solid. All measurements are in metres.

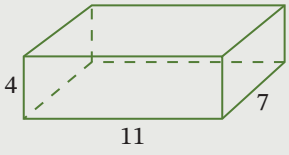
a



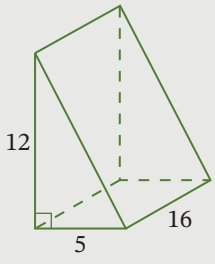
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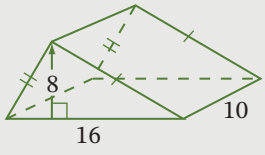
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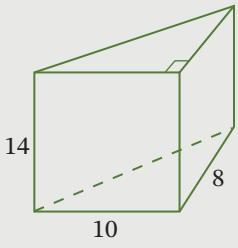
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e



f



Skillsheets
Solid shapes
What is volume?

3.01 Absolute error and percentage error

Video
Decimal places: Photo finish

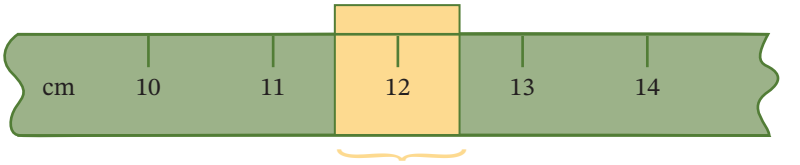
All measurements are approximate. Whether we are using a ruler to measure length, weighing scales for recipe ingredients or a thermometer to measure body temperature, there is some error in every measurement we make.

Absolute error of a measurement

The **absolute error** of a measurement is 0.5 **(half) of the precision** of the measuring instrument's scale.

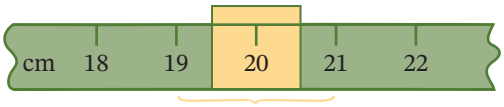
Worksheet
Accuracy in measurement 2

For example, this ruler has a precision of 1 cm, that is, measurements using this ruler are only accurate to the nearest centimetre.



9780170479622

A measurement of 20 cm on this ruler could actually be any size between 19.5 cm and 20.5 cm.



Strictly speaking, it could not be 20.5 cm because that would round up to 21 cm. The range is actually 19.5 cm up to (but not including) 20.5 cm.

Any measurement in the shaded region should be recorded as 20 cm. The measured length is 20 cm, but the actual length is 20 ± 0.5 cm, meaning '20 cm, give or take 0.5 cm' or '19.5 to 20.5 cm'.

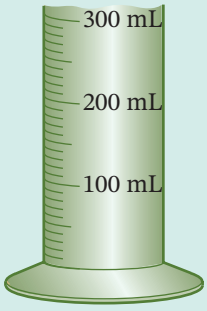
The **absolute error** of this ruler is 0.5 cm, which represents the greatest possible error the actual measurement can differ from the recorded measurement (20 cm), the furthest your measurement could be wrong.

Example 1

Calculate the absolute error of:

a this measuring cylinder

b a measurement of 12.4 cm



Solution

a The cylinder has a precision of 10 mL. — The size of one unit on the scale

Absolute error = 0.5 × 10 mL = 5 mL. — Half of the precision

b A measurement of 12.4 cm has a precision of 0.1 cm. — 12.4 cm is written to the nearest 0.1 cm

Absolute error = 0.5 × 0.1 cm = 0.05 cm.

An absolute error of 1 metre may not be much when measuring the length of a bridge, but it is huge when measuring the length of a bathroom. To understand and compare the sizes of errors when measuring things, we can calculate **percentage error**, the absolute error as a percentage of the measurement.

Percentage error of a measurement

$$\text{Percentage error} = \frac{\text{absolute error}}{\text{measurement}} \times 100\%$$

The smaller the percentage error, the more accurate the measurement, the closer it is to its actual value.

Example 2

The Channel Tunnel is an underwater railway tunnel connecting England and France. It is 50.45 kilometres long.



Gary Perkin/Shutterstock.com

Find the percentage error of this measurement, correct to 2 decimal places.

Solution

Precision = 0.01 km

Absolute error $= 0.5 \times 0.01$ km

$= 0.005$ km

Percentage error $= \frac{0.005}{50.45} \times 100\%$ $\quad \frac{\text{absolute error}}{\text{measurement}} \times 100\%$

$= 0.009\,91...$

$\approx 0.01\ \%$

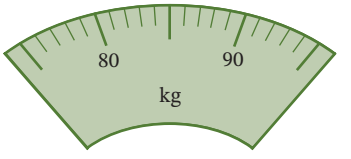
EXERCISE 3.01 ANSWERS ON P. 583

Absolute error and percentage error U F R C

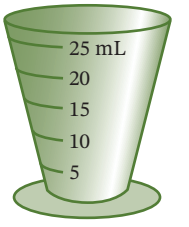
EXAMPLE 1

1 For each measuring instrument, find its precision and absolute error. R C

a



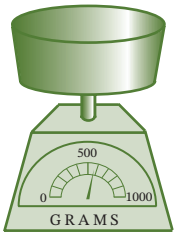
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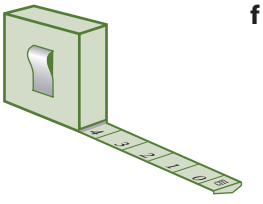
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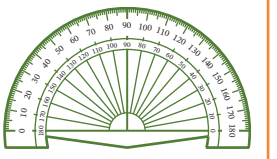
d



e



f



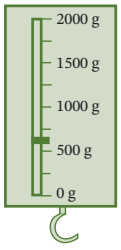
Foundation Standard Complex

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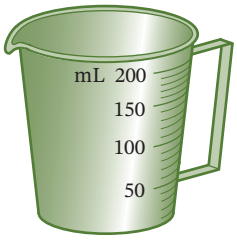
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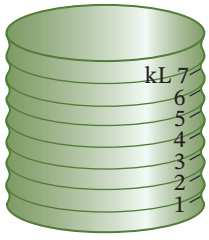
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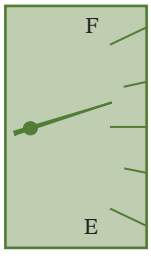
i



j



k



2 Copy and complete this table.

Measurement	Precision	Absolute error
a 58 cm		
b 22.3 mL		
c 31 seconds		
d 175 cm (to the nearest cm)		
e 175 cm (to the nearest 5 cm)		

3 Swara is using a measuring tape marked in millimetres. What is the absolute error of this measurement? Select the correct answer **A**, **B**, **C** or **D**.

A 1 mm

B 0.5 mm

C 1 cm

D 0.5 cm

4 Find the absolute error of each measurement.

a 86 mm

b 2850 L (to the nearest 10 L)

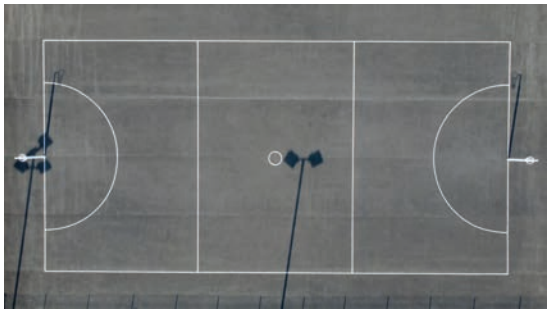
c 9.4 m^2

d 700 kg (to the nearest g)

e 700 kg (to the nearest 10 kg)

f 700 kg (to the nearest 100 kg)

5 A netball court is 15.25 m wide. Calculate the absolute error and hence find the percentage error of this measurement, correct to 3 decimal places.



Richard Newstead/Moment/Getty Images

3.01

EXAMPLE 2

Foundation | Standard | Complex

6 The world's longest river, the Nile river in Egypt, is 6650 km long.

a If this measurement is made to the nearest 10 km, find its absolute error.

b Hence calculate the percentage error, correct to 4 decimal places.



©Frontpage/Shutterstock.com

7 By comparing percentage errors, determine which measurement is Questions 5 and 6 is more accurate, the width of the netball court or the length of the Nile river. R C

8 A timekeeper measures the time it takes an athlete to complete a race to be 32.08 seconds. Calculate the percentage error of this measurement, correct to 4 decimal places.

9 An Olympic marathon is 42.195 km long. Calculate the percentage error of this measurement, correct to 4 decimal places.

10 Which measurement is Questions 8 and 9 is more accurate, the race time or the marathon length? R C

11 *Sweet Juices* is a company using a new machine to bottle its juice. Bottles of juice have a measured capacity with a percentage error of 0.5% of the 500 mL labelled on each bottle. R C

a What is the absolute error of the machine's scale?

b What is the precision of the machine's scale?

c What is the maximum possible capacity?

d What is the minimum possible capacity?

e Which of the following capacities (in mL) could not have been bottled by the machine?

500.05, 498.55, 499.10, 503.10, 502.30, 497.25, 498.50, 502.10

12 The length of a piece of wood is measured to be 8 m. The length of a metal rod is measured to be 8.5 m. By using absolute error in your calculations, show that the piece of wood *could* be longer than the metal rod. R C

13 Why is a measurement of 185.0 cm more accurate than a measurement of 185 cm? R C

14 A cube has a measured length of 4 m. Calculate its maximum possible volume and its minimum possible volume. R C

Rounding error

3.02

Another type of error can occur when performing calculations with measurements, where the measured value or partial answer is rounded too early or severely, resulting in a final answer that is inaccurate (not close to the exact answer). This is called **rounding error**.

Consider the area calculations for purchasing carpet for 3 bedrooms.

Room 1 has dimensions 4.2 m by 3.7 m, Room 2 is 3.9 m by 3.4 m, Room 3 is 3.3 m by 3.3 m.



Pixel-Shot/Adobe Stock

Room 1: $4.2 \times 3.7 = 15.54$ m^2

Room 2: $3.9 \times 3.4 = 13.26$ m^2

Room 3: $3.3 \times 3.3 = 10.89$ m^2

Total area $= 39.69$ m^2

If we were to estimate by rounding measurements to the nearest metre, then:

Room 1: $4 \times 4 = 16$ m^2

Room 2: $4 \times 3 = 12$ m^2

Room 3: $3 \times 3 = 9$ m^2

Total area $= 37$ m^2

Notice that the estimate is too small, and if we had rounded the measurements we would not have bought enough carpet.

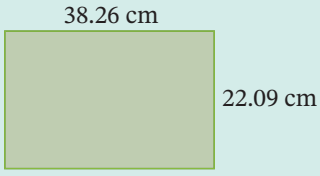
Rounding during calculations can lead to errors in solutions. If approximate values are used in further calculations, the error becomes worse.

The table below shows a number being **rounded** and **truncated** to the nearest whole number, to one decimal place, to 3 decimal places and to 5 decimal places. **Truncate** means to 'cut off' the number at the decimal place regardless of the next digit (or 'rounding down').

10.638 971 564 538 ...	Rounded	Truncated
to the nearest whole number	11	10
to one decimal place	10.6	10.6
to 3 decimal places	10.639	10.638
to 5 decimal places	10.638 97	10.638 97

Example 3

A rectangle has length 38.26 cm and width 22.09 cm.



a Calculate its **perimeter** and area.

b Round the rectangle's length and width to one decimal place and then calculate its perimeter and area with these values and then find correct to one decimal place the percentage error of the perimeter and area.

c Truncate the rectangle's length and width to the nearest whole number, then calculate its perimeter and area with these values, then find correct to one decimal place the percentage error of the perimeter and area.

Solution

a Perimeter $= 2 \times 38.26 + 2 \times 22.09$
$= 120.7$ cm

Area $= 38.26 \times 22.09$
$= 845.1634$ cm^2

b Length $= 38.3$ cm, width $= 22.1$ cm

Perimeter $= 2 \times 38.3 + 2 \times 22.1$
$= 120.8$ cm

Error $= 120.8 - 120.7$
$= 0.1$ cm

Percentage error $= \frac{0.1}{120.7} \times 100\%$
$= 0.0828 \ldots$
$\approx 0.1\%$

Area $= 38.3 \times 22.1$
$= 846.43$ cm^2

Error $= 846.43 - 845.1634$
$= 1.2666$

Percentage error $= \frac{1.2666}{845.1634} \times 100\%$
$= 0.1498 \ldots$
$\approx 0.1\%$

c Length $= 38$ cm, width $= 22$ cm

Perimeter $= 2 \times 38 + 2 \times 22$
$= 120$ cm

Error $= 120.7 - 120$
$= 0.7$ cm

Percentage error $= \frac{0.7}{120.7} \times 100\%$
$= 0.5799 \ldots$
$\approx 0.6\%$

Area $= 38 \times 22$
$= 836$ cm^2

Error $= 845.1634 - 836$
$= 9.1634$

Percentage error $= \frac{9.1634}{845.1634} \times 100\%$
$= 1.0842$
$\approx 1.1\%$

- Notice that the percentage error is higher for area than it is for perimeter: this is due to addition being used to calculate perimeter, whereas by using multiplication to calculate areas, the error is being multiplied.
- Also notice that the percentage error is higher the more we round or truncate the measurements or values. When calculating, it is best to use the exact or most accurate measurements to minimise any rounding errors.

EXERCISE 3.02 ANSWERS ON P. 583

Rounding error U F R C

1 Copy and complete this table to round and truncate the value of π: 3.141 592 6 ...

3.141 592 6 ...	Rounded	Truncated
a to the nearest whole number		
b to 2 decimal places		
c to 4 decimal places		
d to 6 decimal places		

EXAMPLE 3

2 A rectangle has length 6.49 m and width 3.82 m.

a Calculate its area.

b Truncate each measurement to one decimal place, then calculate its approximate area.

c Round each measurement to the nearest whole number, then calculate its approximate area.

3 A circle has radius 7.63 cm.

a Calculate its area correct to 2 decimal places.

b Calculate its area correct to 2 decimal places, using $\pi = 3.14$, then find correct to 2 decimal places the percentage error of the calculated area.

c Round the circle's radius to one decimal place, then calculate its area correct to 2 decimal places, then find correct to 2 decimal places the percentage error of the calculated area.

4 A square has length 11.27 m. R C

a Calculate its perimeter and area.

b Truncate the square's length to one decimal place and use this value to calculate its perimeter and area, then find to one decimal place, the percentage error of the calculated perimeter and area.

c Round the square's length to the nearest whole number and use this value to calculate its perimeter and area, then find to one decimal place the percentage error of the perimeter and area.

5 A rectangle has dimensions 8.25 cm by 6.09 cm. R C

a Calculate its perimeter and area.

b Truncate the rectangle's length and width to one decimal place, then calculate its perimeter and area and find to one decimal place the percentage error of the calculated perimeter and area.

c Round the rectangle's length and width to the nearest whole number, then calculate its perimeter and area and find to one decimal place the percentage error of the perimeter and area.

6 A circle has diameter 8 m, measured correct to the nearest metre. R C

- **a** Calculate the circumference of the circle, correct to 4 decimal places.
- **b** Find the absolute error of the diameter.
- **c** Hence, calculate:
  - **i** correct to 4 decimal places, the smallest possible circumference of the circle
  - **ii** correct to 2 decimal places, the percentage error of the smallest possible circumference
  - **iii** correct to 4 decimal places, the largest possible circumference
  - **iv** correct to 2 decimal places, the percentage error of the largest possible circumference.

7 A right-angled triangle has a base length of 60 mm and a perpendicular height of 90 mm, measured to the nearest mm. R C

- **a** Calculate the length of the hypotenuse, correct to 4 decimal places.
- **b** Find the absolute error of the base length and perpendicular height.
- **c** Hence calculate correct to 4 decimal places the minimum possible length of its hypotenuse and correct to one decimal place its percentage error.
- **d** Calculate the area of this triangle.
- **e** Hence calculate the maximum possible area of the triangle and correct to one decimal place its percentage error.

8 A rectangular swimming pool has side lengths of 3 m and 8 m, correct to the nearest metre. A fence needs to be placed 1 m out from the edge of the pool enclosing the entire pool. Calculate the maximum and minimum perimeter of this fence. R

3.03 Areas of composite shapes

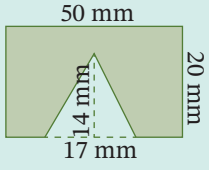
Puzzle
Area

Worksheet
A page of circular shapes

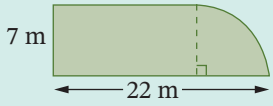
Example 4

Find the area of each composite shape, correct to one decimal place where appropriate.

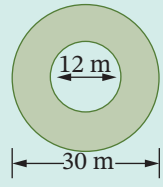
a



b



c



Solution

a $\text{Area} = 50 \times 20 - \frac{1}{2} \times 17 \times 14$
$= 881 \text{ mm}^2$

Area of rectangle − area of triangle

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b The shape is made up of a rectangle and a quadrant.

Radius of quadrant $= 7$ m

Length of rectangle $= 22 - 7$

$= 15$ m

Area of shape = area of rectangle + quadrant

$= 15 \times 7 + \frac{1}{4} \times \pi \times 7^2$

$= 143.4845...$

≈ 143.5 m^2

c This ring shape is an **annulus**, its area is enclosed by 2 circles with the same centre.

Radius of large circle $= \frac{1}{2} \times 30$ m

$= 15$ m

Radius of small circle $= \frac{1}{2} \times 12$ m

$= 6$ m

Area of annulus $= \pi \times 15^2 - \pi \times 6^2$ Large circle − small circle

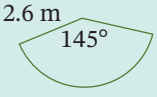
$= 593.7610....$

$= 593.8$ m^2

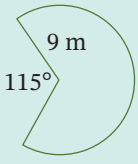
Example 5

Calculate, correct to 2 decimal places, the area of each sector.

a



b



A **sector** is a fraction of a circle 'cut' along 2 radii, like a pizza slice.

Solution

a Area $= \frac{145}{360} \times \pi \times 2.6^2$ $\frac{145}{360} \times$ area of circle

$= 8.55385...$

≈ 8.55 m^2

There are 360° in a circle, but a sector is a fraction of a circle

b Sector angle $= 360° - 115°$

$= 245°$

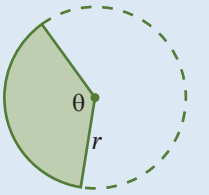
Area of sector $= \frac{245}{360} \times \pi \times 9^2$

$= 173.18029...$

≈ 173.18 m^2

Area of a sector

$$\text{Area of a sector} = \frac{\theta}{360} \times \pi r^2$$



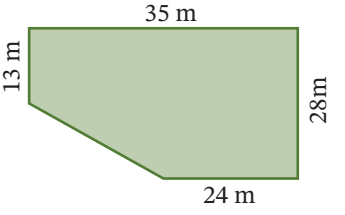
EXERCISE 3.03 ANSWERS ON P. 583

Area of composite shapes U F PS R

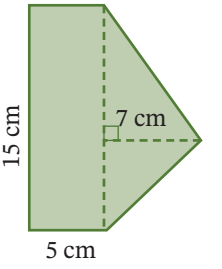
EXAMPLE 4

1 Find the area of each composite shape. PS R

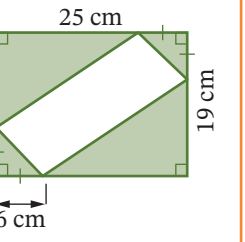
a



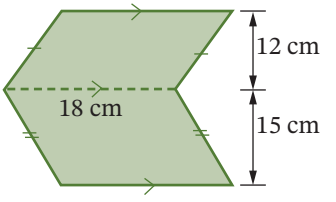
b



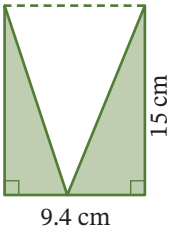
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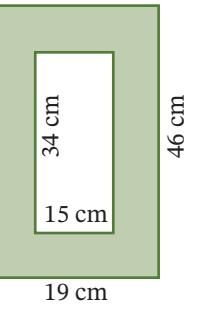
d



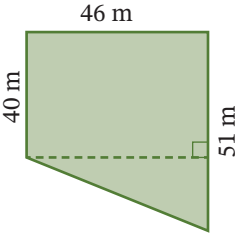
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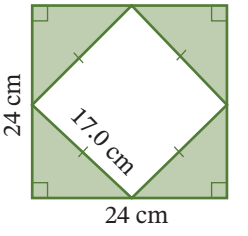
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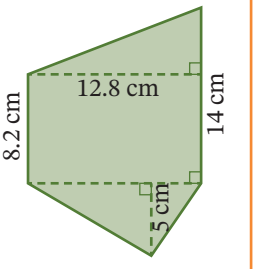
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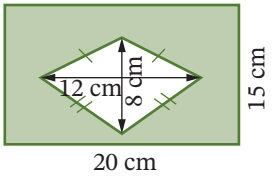
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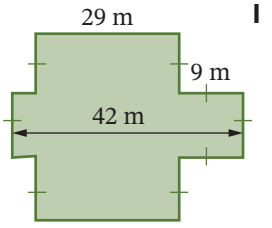
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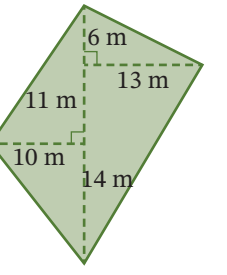
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k



l

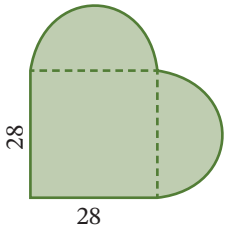


Foundation Standard Complex

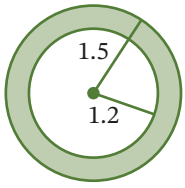
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2 Calculate, correct to one decimal place, the area of each shape. All measurements are in metres. PS R

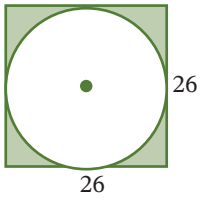
a



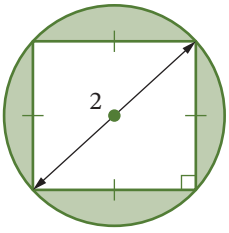
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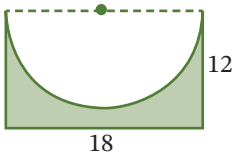
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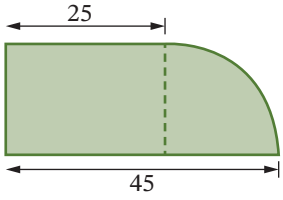
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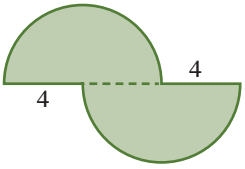
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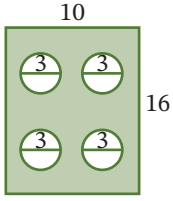
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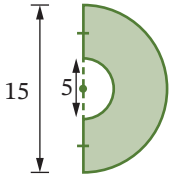
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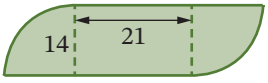
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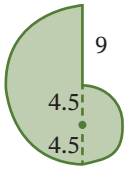
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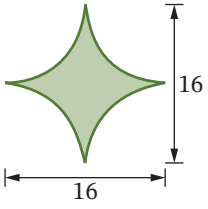
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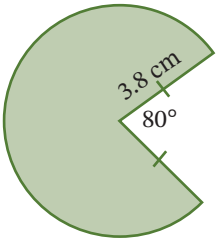
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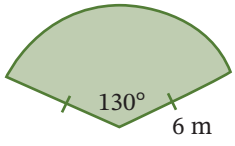
3 Find, correct to one decimal place, the area of each sector.

EXAMPLE 5

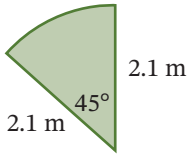
a



b



c



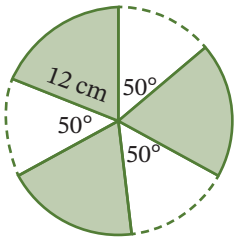
4 A bike tyre has a diameter of 715 mm.

a How far will the bike travel in one revolution of the tyre? Give your answer in metres, correct to 2 decimal places.

b How many revolutions of the tyre are required to travel a distance of 5 km?

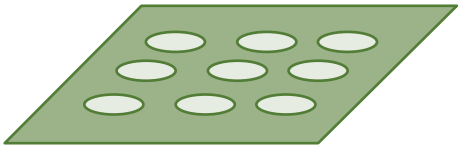
Foundation Standard Complex

5 Calculate the area of the shaded region. Select the correct answer **A**, **B**, **C** or **D**. R



A 362.7 cm²
B 452.4 cm²
C 188.5 cm²
D 263.9 cm²

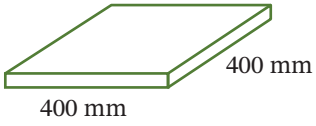
6 A rectangular metal plate with dimensions 2.5 m × 2.2 m has 9 holes of diameter 46 cm drilled in it.



a Find the total area of the holes that have been drilled. Give your answer in m², correct to 2 decimal places.

b What percentage of the metal plate remains?

7 A rectangular courtyard 15 m long and 8 m wide is to be covered with square pavers of side length 400 mm, costing $79.29/m². PS R



a What is the area of one paver, in m²?

b How many pavers will be required to pave the courtyard?

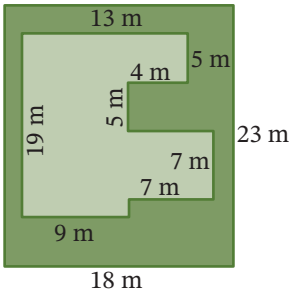
c Calculate the cost of paving the courtyard.

8 A circular sports ground of diameter 140 m has a rectangular soccer pitch measuring 110 m by 70 m inside it. The area outside the soccer pitch is to be painted in the team colour of red.

a Calculate the area that is to be painted red, correct to the nearest m².

b If the cost of paint is $29.50 per 50 m², calculate the cost of painting this area.

9 The diagram shows the floor plan of a house on a block of land. PS R



a Calculate the area of the block.

b Calculate the area taken up by the house.

c What percentage of the area of the block is taken up by the house?

d The area not covered by the house and is to be turfed. Find the cost of turfing the yard at a cost of $11.75/m².

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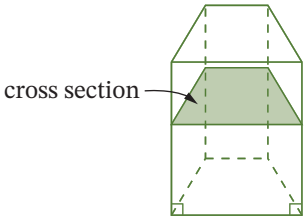
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Surface area of a prism

3.04

A **cross-section** of a solid is a 'slice' of the solid cut *across* it, parallel to its end faces, rather than along it. A **prism** has the same (uniform) cross-section along its length, and each cross-section is a **polygon** (with straight sides).

The trapezoidal prism shown here has cross-sections that are trapeziums.



A right prism

Worksheets
Surface area 2
Nets of solids

3.04

Surface area

The **surface area** of a solid is the total area of all the faces of the solid. To calculate the surface area of a solid, find the area of each face and add the areas together.

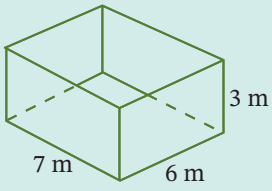
Videos
Surface area of a prism
Surface areas of prisms

It is often useful to draw the net of a solid when finding its surface area. A net may be used to form an open solid or a closed solid. A sealed cardboard box is an example of a **closed solid**. A cardboard box with the lid removed is an example of an **open solid**. For the surface area of an open solid, we only count the *external* surfaces, not the internal ones (so that we don't count each surface twice).

Example 6

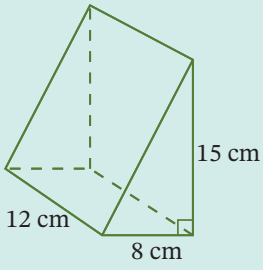
Find the surface area of each prism.

a



Open rectangular prism

b



Closed triangular prism

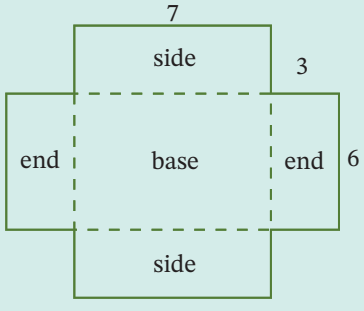
Solution

a This open prism has 5 faces.

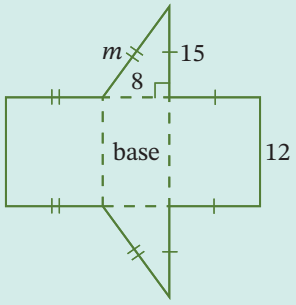
Surface area = 2 ends + 2 sides + base

$= 2 \times (3 \times 6) + 2 \times (3 \times 7) + (6 \times 7)$

$= 120 \text{ m}^2$



b This closed prism has 5 faces: 2 identical triangles (front and back) and 3 different rectangles.



Using Pythagoras' theorem to find m, the hypotenuse of the triangle:

$m^2 = 8^2 + 15^2$

$= 289$

$m = \sqrt{289}$

$= 17$

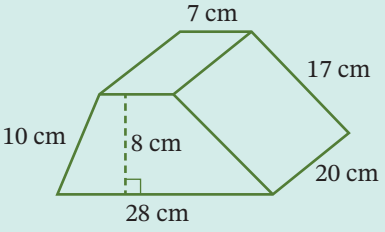
Surface area = 2 triangles + 3 rectangles

$$= 2 \times \left(\frac{1}{2} \times 8 \times 15 \right) + (17 \times 12) + (8 \times 12) + (15 \times 12)$$

$= 600 \text{ cm}^2$

Example 7

Calculate the surface area of this trapezoidal prism.



Solution

This trapezoidal prism has 6 faces:

2 identical trapeziums (front and back) and 4 different rectangles.

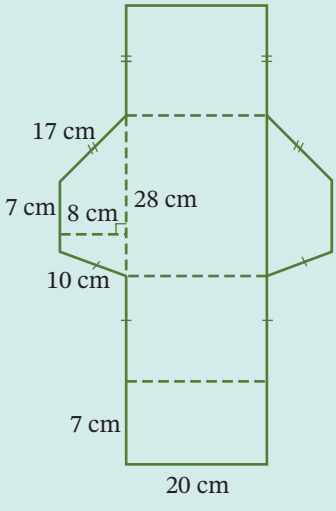
Area of each trapezium $= \frac{1}{2} \times (7 + 28) \times 8$

$= 140 \text{ cm}^2$

Surface area $= (2 \times 140) + (20 \times 7) + (20 \times 10)$

$+ (20 \times 28) + (20 \times 17)$

$= 1520 \text{ cm}^2$



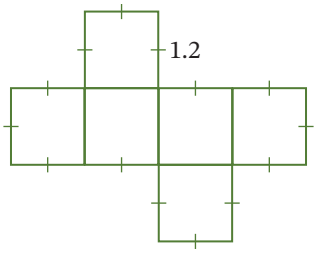
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EXERCISE 3.04 ANSWERS ON P. 583

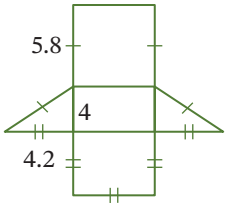
Surface area of a prism U F PS R C

1 Identify the prism that each net represents, then calculate the surface area of the prism. All lengths are in metres. R C

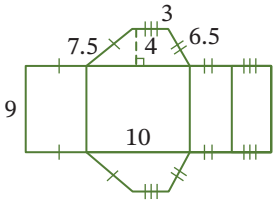
a



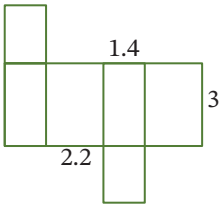
b



c



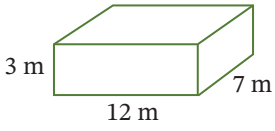
d



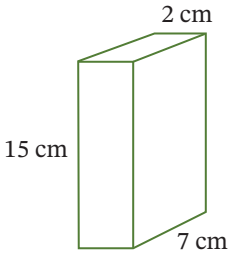
2 Find the surface area of each prism.

EXAMPLE 6

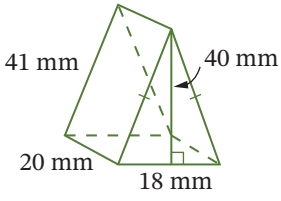
a



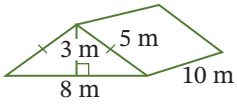
b



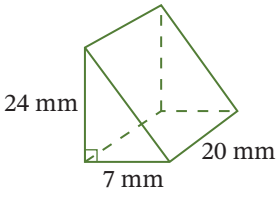
c



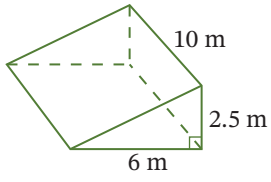
d



e

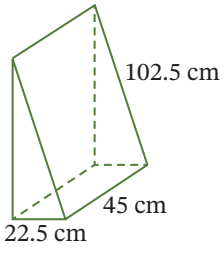


f



3 Calculate the surface area of this triangular prism.
Select the correct answer **A**, **B**, **C** or **D**.

A 12 375 cm²
B 11 250 cm²
C 10 125 cm²
D 12 431.25 cm²

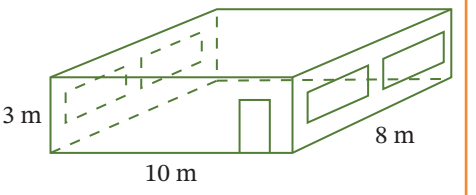


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4 This classroom is being renovated. Find: PS

a the area of the floor to be carpeted and the total cost, at $55 per square metre.

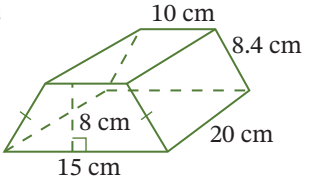
b the ceiling and wall area to be painted, if the room contains 4 windows, each 2.5 m by 1.5 m, and a doorway 2 m by 0.8 m.



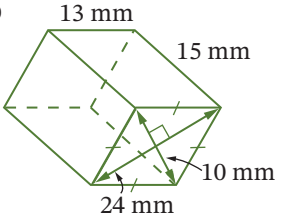
EXAMPLE 7

5 Calculate the surface area of each prism.

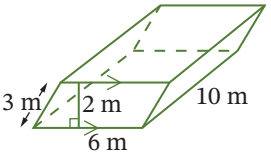
a



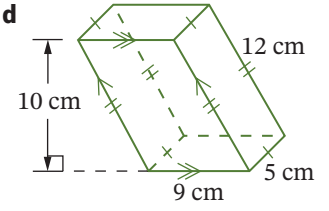
b



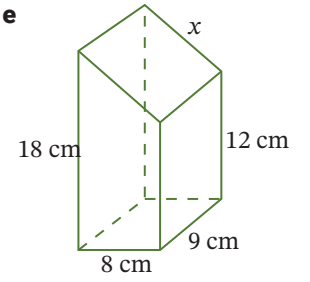
c



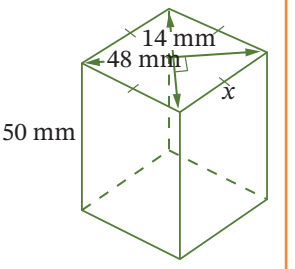
d



e



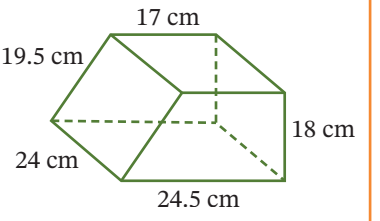
f



6 Calculate the surface area the trapezoidal prism. Select **A**, **B**, **C** or **D**.

A 10 584 cm^2 **B** 2643 cm^2

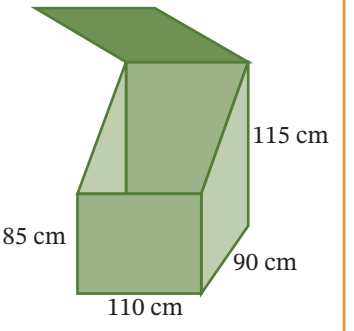
C 2082.75 cm^2 **D** 8964 cm^2



7 The wooden toy box is in the shape of a trapezoidal prism.

a Calculate how much timber is required to make the toy box, correct to the nearest cm^2.

b If the price of the timber is $25/m^2, calculate the cost of making the box.



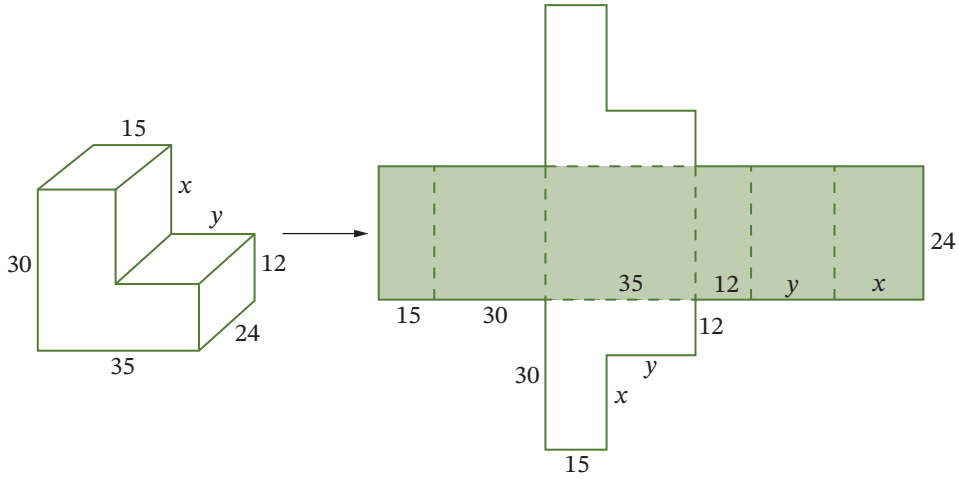
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Investigation

A surface area shortcut

1 Consider this L-shaped prism and its net. We will find its surface area.



a Find x and y.

b This prism has 8 faces: 2 'L-shaped' ends and 6 rectangles. Instead of calculating the areas of the 6 rectangles separately, we can combine them into one long rectangle, as shaded in the net above. The length of the rectangle is the same as the perimeter of the L-shape. What is the length of this long rectangle?

c What is the area of this long rectangle?

d Copy and complete:

Length of shaded rectangle = p _____ of the L-shape.

e Find the surface area of the prism by copying and completing the following:

Surface area = 2 'L-shaped' ends + shaded rectangle

$= 2 \times (15 \times 30 + 20 \times 12) +$ _____

$=$ _____

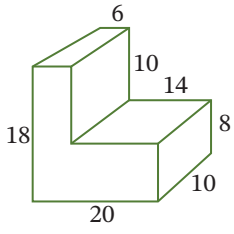
2 From question **1**, it can be seen that the surface area of any prism with end faces of area A and perpendicular height (distance between end faces) h can be calculated using the formula:

$$SA = 2A + Ph$$

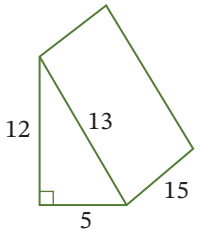
where P = perimeter of the end face.

Use this method to calculate the surface area of each prism. All measurements are in centimetres.

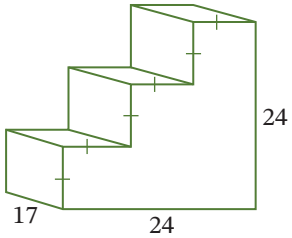
a



b



c

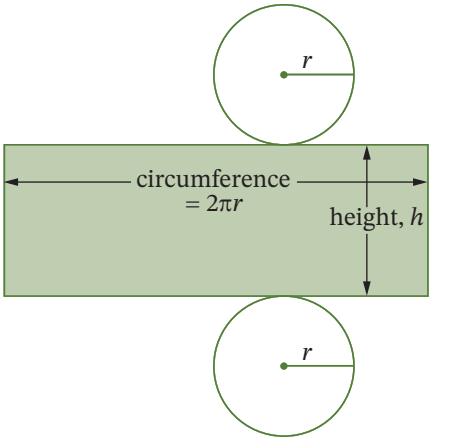


3.05 Surface area of a cylinder

Puzzle
Surface area

Video
Surface area of a cylinder

A closed cylinder has 3 faces made up of 2 circles (the circular ends) and a rectangle (the curved surface). The length of the rectangle is the **circumference** of the circular end, while the width of the rectangle is the height of the cylinder.



Surface area of a cylinder = area of 2 circles
+ area of rectangle

$$SA = 2 \times \pi r^2 + 2\pi r \times h$$
$$= 2\pi r^2 + 2\pi rh$$

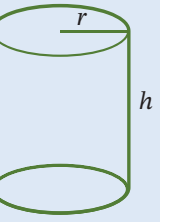
Surface area of a closed cylinder

$$SA = 2\pi r^2 + 2\pi rh$$

where r = radius of circular base

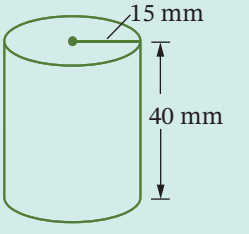
h = perpendicular height

The area of the 2 circular ends = $2\pi r^2$
and the area of the curved surface = $2\pi rh$.



Example 8

Find, correct to the nearest mm², the surface area of this cylinder.



Solution

Surface area = area of 2 ends + area of the curved surface $\quad r = 15, h = 40$

$$= 2\pi r^2 + 2\pi rh$$
$$= 2 \times \pi \times 15^2 + 2 \times \pi \times 15 \times 40$$
$$= 5183.627...$$
$$\approx 5184 \text{ mm}^2$$

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Example 9

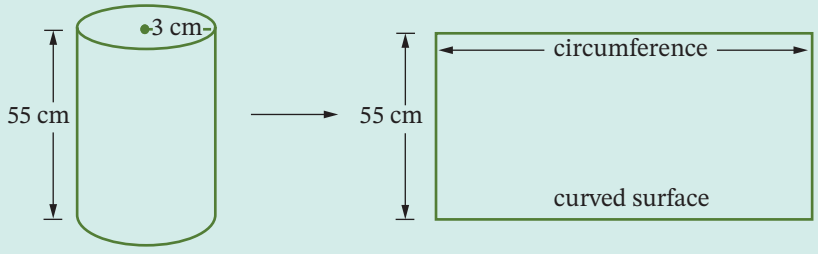
Find, correct to 2 **significant figures**, the surface area of:

a a cylindrical tube, open at both ends, with radius 3 cm and height 55 cm

b an open half-cylinder with radius 0.65 m and length 2.4 m.

Solution

a



Surface area = curved surface

$= 2\pi rh$ $\quad r = 3$ and $h = 55$

$= 2 \times \pi \times 3 \times 55$

$= 1036.725...$

$= 1000 \text{ cm}^2$

b



Surface area = 2 semicircle ends $+ \frac{1}{2} \times$ curved surface

$= 2 \times \left(\frac{1}{2} \times \pi \times 0.65^2 \right) + \frac{1}{2} \times (2 \times \pi \times 0.65 \times 2.4)$

$= 6.2282...$

$\approx 6.2 \text{ m}^2$

EXERCISE 3.05 ANSWERS ON P. 583

Surface area of a cylinder U F PS R

EXAMPLE 8

1 Calculate, correct to one decimal place, the surface area of a cylinder with:

a radius 1.4 m, height 2.2 m

b diameter 45 cm, height 65 cm

c diameter 9 cm, height 24 cm

d radius 1.3 m, height 3.8 m

2 Find, correct to the nearest whole number, the curved surface area of a cylinder with:

a radius 1.5 m, height 3.75 m

b diameter 27 cm, height 41 cm

3 A container of potato chips is a cylinder with diameter 7 cm and height 23.2 cm. Calculate its surface area, correct to one decimal place.

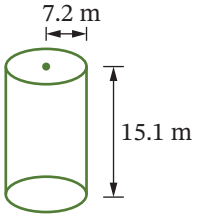
EXAMPLE 9

4 Find the surface area of a cylinder, open at one end with diameter 12 mm and length 15 cm. Select the closest answer **A**, **B**, **C** or **D**. R

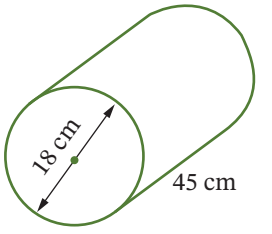
A 678.6 cm^2 **B** 1017 cm^2 **C** 6107.3 mm^2 **D** 5768.0 mm^2

5 Calculate, correct to the nearest whole number, the surface area of each solid. R

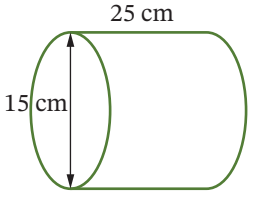
a closed cylinder



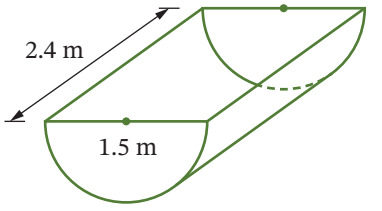
b closed cylinder



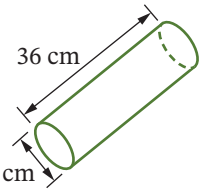
c cylinder with one open end



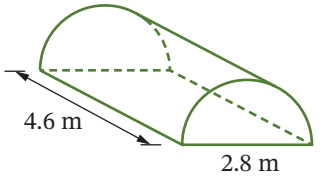
d half cylinder with open top



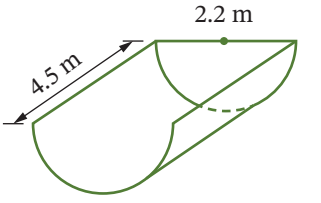
e cylinder open both ends



f closed half cylinder



g half cylinder with open top, one end open

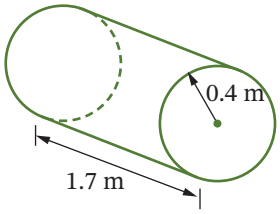




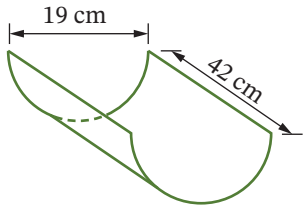
Foundation Standard Complex

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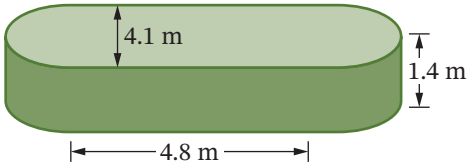
h closed cylinder



i half cylinder, open both ends



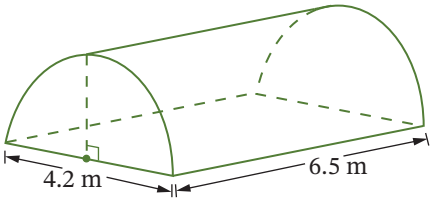
6 The inside of the swimming pool, including the floor, is to be repainted. Find: PS R



a the area to be repainted, correct to one decimal place

b the number of whole litres of paint needed if coverage is 9 m^2 per litre.

7 The diagram shows a tent to be made in the shape of half a cylinder. Find: PS R



a the area of the floor of the tent

b the surface area of the tent, excluding the floor.

c the total cost of materials for the tent if the material for the flooring costs \$18.50/m^2 and the canvas for the tent costs \$21.75/m^2.



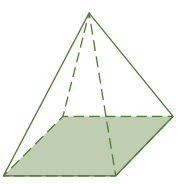
Alamy Stock Photo/Robert Wyatt

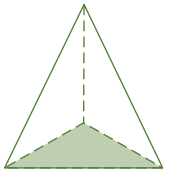
Foundation Standard Complex

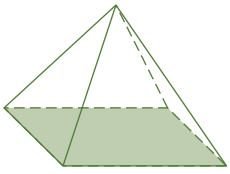
3.06 Surface area of a pyramid

PATH

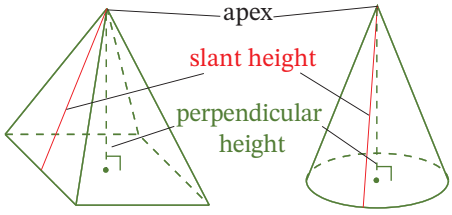
A **pyramid** is a solid shape with a polygon for its base and triangular faces that meet at a point or vertex called its **apex**. Like a prism, a pyramid is named by the shape of its base.


Square pyramid


Triangular pyramid


Rectangular pyramid

A **cone** is a solid shape with a circular base and a curved surface that also has an **apex**. However, a cone is not a pyramid because its base is not a polygon (a circle does not have straight sides).



The **slant height** of a pyramid or cone is the height from the apex to the base, along a side face. It is different from the **perpendicular height** of a pyramid or cone, which is the perpendicular distance from the apex to the base.

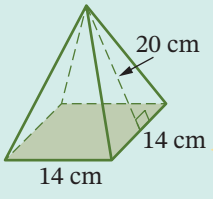
The surface area of a pyramid is calculated by adding the areas of its base and its triangular faces.

Video
Surface area of a pyramid

Example 10

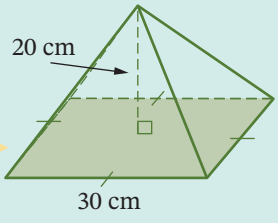
Find the surface area of each square pyramid.

a



This pyramid has a slant height of 20 cm

b



This pyramid has a perpendicular height of 20 cm

Solution

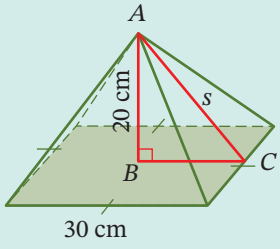
a Surface area = area of square base + area of 4 triangular faces

$$= 14 \times 14 + 4 \times \frac{1}{2} \times 14 \times 20$$
$$= 756 \text{ cm}^2$$

b First find the slant height, s, using Pythagoras' theorem.

In $\triangle ABC$, $s^2 = 20^2 + 15^2$ $\quad BC = \frac{1}{2} \times 30$

$$= 625$$
$$s = \sqrt{625}$$
$$= 25 \text{ cm}$$



Surface area $= 30 \times 30 + 4 \times \frac{1}{2} \times 30 \times 25$

$$= 2400 \text{ cm}^2$$

9780170479622

Example 11

A rectangular pyramid with base 10 cm by 8 cm has a perpendicular height of 15 cm. Find its surface area, correct to one decimal place.

Solution

First find the slant heights AP and AQ.

$AP^2 = AX^2 + XP^2$

$= 15^2 + 4^2$ $\quad XP = \frac{1}{2} \times 8$

$= 241$

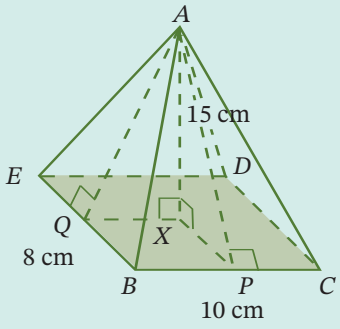
$AP = \sqrt{241}$ cm

$AQ^2 = AX^2 + XQ^2$

$= 15^2 + 5^2$ $\quad XQ = \frac{1}{2} \times 10$

$= 250$

$AQ = \sqrt{250}$ cm



It is better to leave the lengths of AP and AQ in surd form rather than round them to decimals so that the final answer is accurate.

Surface area = area of rectangle base + 2 × Area $\triangle ABC$ + 2 × Area $\triangle ADC$

$= 10 \times 8 + 2 \times \frac{1}{2} \times 10 \times \sqrt{241} + 2 \times \frac{1}{2} \times 8 \times \sqrt{250}$

$= 361.7328...$

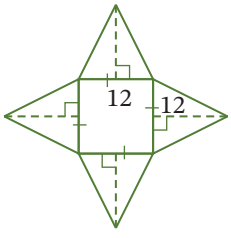
≈ 361.7 cm^2

EXERCISE 3.06 ANSWERS ON P. 584

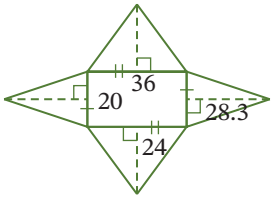
Surface area of a pyramid U F PS R

1 Find the area of each net and hence the surface area of the corresponding pyramid. All measurements are in centimetres.

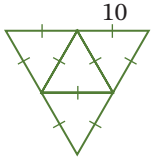
a



b



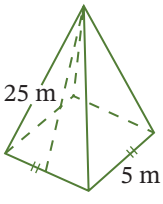
c



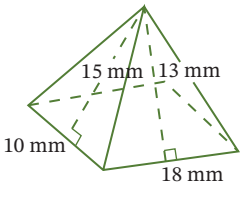
2 Find the surface area of each pyramid.

EXAMPLE 10

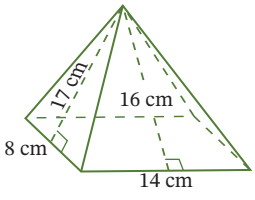
a



b



c

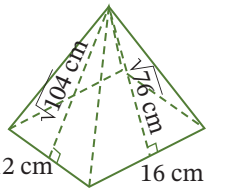


Foundation | Standard | Complex

PATH

3 Find the surface area of this pyramid.
Select the closest answer **A**, **B**, **C** or **D**.

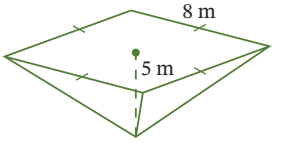
A 288.0 cm²
B 453.9 cm²
C 459.8 cm²
D 471.0 cm²



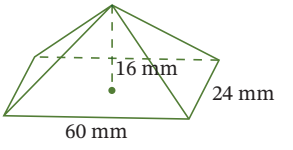
EXAMPLE 11

4 Calculate, correct to one decimal place, the surface area of each pyramid. R

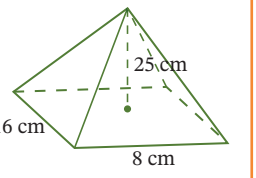
a



b

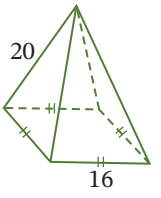


c

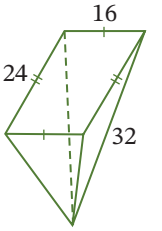


5 Calculate, correct to the nearest square centimetre, the surface area of each pyramid.
All measurements are in centimetres. PS R

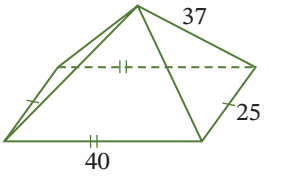
a



b



c



6 The great pyramid of Khufu (or Cheops) in Egypt has a height of 147 m, and each side of its square base measures 230 m. Find its surface area (excluding the base), correct to the nearest square metre. PS R

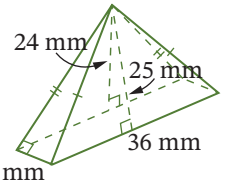
7 The Louvre Art Museum in Paris, France, has a large pyramid at its entrance (built in 1989). It reaches a height of 21.6, has a square base of length 34 m and its sides are covered in rhombus and triangular glass pieces. Calculate the surface area of the glass pyramid (to the nearest m³). PS R



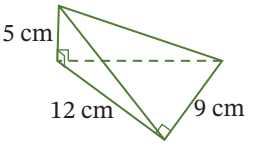
vichie81/Shutterstock.com

8 Calculate, correct to one decimal place, the surface area of each pyramid. PS R

a

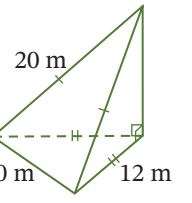


b



All faces are right-angled triangles

c



9 A square pyramid has a surface area of 4704 m² and a base area of 1764 m². Find: R

a the length of its base
b the area of each triangular face
c the slant height of each triangular face
d the perpendicular height of the pyramid.

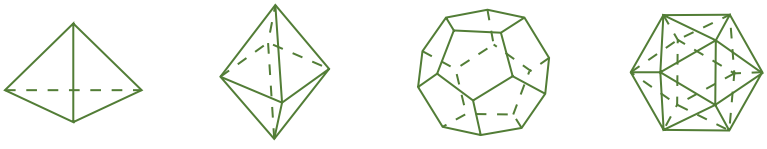
Foundation Standard Complex

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Did you know?

The Platonic solids

The cube is called a **regular polyhedron** or **Platonic solid** because it has 6 identical faces. The more formal name for a cube is **regular hexahedron**, 'hex' meaning 6. There are only 6 possible regular polyhedrons: the other 5 are shown below. Each face on a regular polyhedron is a **regular polygon.** The regular tetrahedron, for example, is a pyramid of equilateral triangles.



Regular tetrahedron
4 faces

Regular octahedron
8 faces

Regular dodecahedron
12 faces

Regular icosahedron
20 faces

The tetrahedron, cube and octahedron occur in nature in the form of certain crystals.

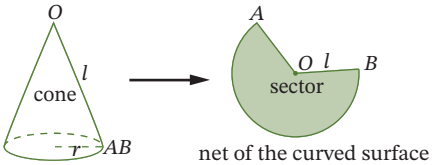
What are the shapes of the faces of each Platonic solid shown?

Investigation

PATH

The surface area of a cone

The net of a cone is made up of a circle (for the base) and a sector of a circle (for the curved surface). The second diagram below shows the curved surface of a cone.



cone

net of the curved surface of the cone

We can use this fact to find a formula for the curved surface area of a cone. Suppose the cone has base radius r and slant height l. Then the sector must have radius l.

Looking at the sector, the major arc length AB is a fraction of the circumference of the circle and the area of the sector is a fraction of the area of the circle. They should be the same fraction, so:

$$\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{\text{Arc length}}{\text{Circumference}}$$

1 The major arc length AB is the same as the circumference AB of the base of the cone in the first diagram. Write an algebraic expression for the circumference of the circle in the first diagram.

2 Write an algebraic expression for the circumference of the complete circle in the second diagram.

3 Write an algebraic expression for the *area* of the complete circle in the second diagram.

4 $\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{\text{Arc length}}{\text{Circumference}}$ becomes $\frac{\text{Area of sector}}{\pi l^2} = \frac{2\pi r}{2\pi l}$

Copy and complete: $\therefore$ Area of sector $= \frac{2\pi r}{2\pi l} \times \pi l^2 =$ _____

PATH

5 But the area of the sector is equal to the curved surface area of the cone.

Copy and complete:

Surface area of cone = area of circular base + area of curved surface

= _____ + _____

3.07 Surface areas of cones and spheres

PATH

Surface area of a cone

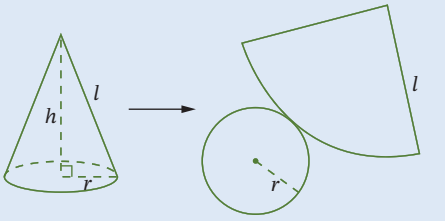
$$SA = \pi r^2 + \pi rl$$

where r = radius of circular base

l = slant height

The area of the circular base = πr^2

and the area of the curved surface = πrl

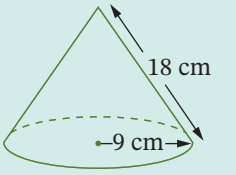


Example 12

For this cone, find correct to one decimal place:

a the curved surface area

b the total surface area.



Solution

$r = 9$ cm and $l = 18$ cm

a Curved surface area = πrl

$= \pi \times 9 \times 18$

$= 508.9380...$

≈ 508.9 cm^2

b Total surface area = $\pi r^2 + \pi rl$

$= \pi \times 9^2 + \pi \times 9 \times 18$

$= 763.4070...$

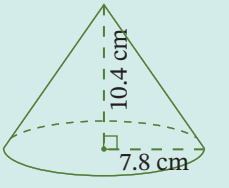
≈ 763.4 cm^2

Videos
Surface area of a cone and sphere

Surface area and volume of a cone

Example 13

Find, correct to 2 decimal places, the surface area of this cone.



Solution

First calculate the slant height:

$l^2 = 7.8^2 + 10.4^2$

$= 169$

$l = \sqrt{169}$

$= 13$

Surface area = $\pi r^2 + \pi rl$

$= \pi \times 7.8^2 + \pi \times 7.8 \times 13$

$= 509.6919...$

≈ 509.69 cm^2

A **sphere** is a ball shape, completely round.
A **hemisphere** is half a sphere.

PATH

3.07

Surface area of a sphere

$SA = 4\pi r^2$

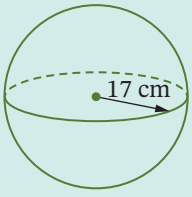
where r = radius of the sphere



Note that the surface area of a sphere is 4 times the area of a circle that shares its centre.

Example 14

Find, correct to 2 decimal places, the surface area of this sphere.



Video
Surface area of a cone and sphere

Solution

Surface area $= 4\pi r^2$

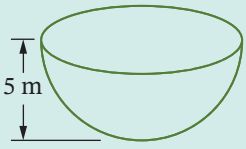
$= 4 \times \pi \times 17^2$

$= 3631.6811...$

$= 3631.68 \text{ cm}^2$

Example 15

Find the surface area of this closed hemisphere in exact form, in terms of π.



Solution

Surface area = Area of circular base + Curved surface area

$= \pi r^2 + \frac{1}{2} \times 4\pi r^2$

$= \pi r^2 + 2\pi r^2$

$= 3\pi r^2$

$= 3 \times \pi \times 5^2$

$= 75\pi \text{ m}^2$

Answers written as surds or in terms of π are exact because they are not decimal approximations.

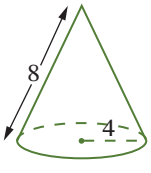
EXERCISE 3.07 ANSWERS ON P. 584

Surface areas of cones and spheres U F PS R C

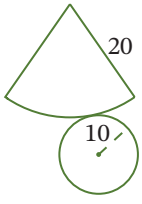
EXAMPLE 12

1 Calculate, correct to the nearest cm^2, the curved surface area of each cone.
All measurements are in centimetres.

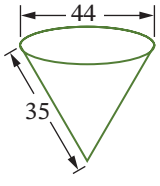
a



b



c

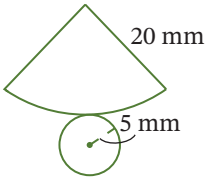


Foundation | Standard | Complex

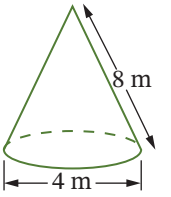
PATH

2 Find, correct to one decimal place, the total surface area of each cone.

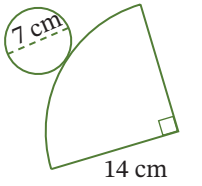
a



b



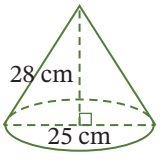
c



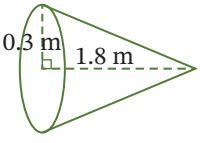
EXAMPLE 13

3 Calculate, correct to one decimal place, the surface area of each cone. R

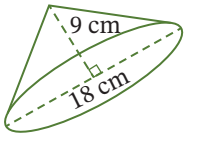
a



b

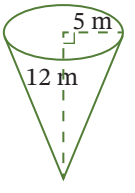


c

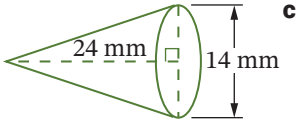


4 Find in exact form (in terms of π) the total surface area of each cone. R C

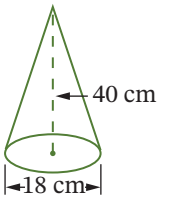
a



b



c



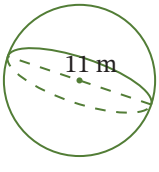
EXAMPLE 14

5 Calculate, correct to one decimal place, the surface area of each sphere.

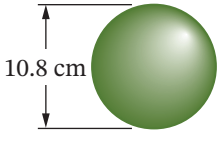
a



b



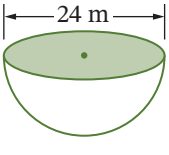
c



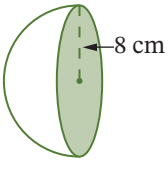
EXAMPLE 15

6 Find in exact form the surface area of each closed hemisphere. R C

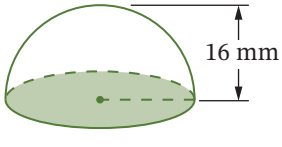
a



b



c



7 Calculate the surface area of a closed hemisphere with radius 1.4 m.
Select the closest answer **A**, **B**, **C** or **D**. R

A 12.32 m² **B** 18.47 m² **C** 24.63 m² **D** 49.26 m²

8 Find, correct to the nearest square metre, the surface area of: PS R

a a sphere with diameter 10 m

b an open cone with base radius 10 m, slant height 20 m

c an open hemisphere with radius 10 m

d a cone with base diameter 10 m, perpendicular height 20 m.

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PATH

3.07

9 The Earth has a radius of approximately 6378 km. PS R C

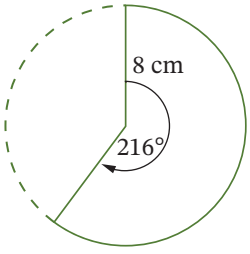
a Calculate the surface area of the Earth in scientific notation, correct to 3 significant figures.

b If 70.9% of the Earth's surface is covered by water, calculate the land area of the Earth's surface, correct to 3 significant figures.

c The land area of Australia is 7 692 000 km^2. Calculate Australia's land area as a percentage of the Earth's land area (correct to 2 decimal places).

10 Find the amount of sheet metal needed to form a conical funnel of base radius 30 cm and vertical height 50 cm, allowing for a 0.5 cm overlap at the join. PS R C

11 A cone is made from a sector of a circle with radius 8 cm and central angle 216°. Find: PS R C



a the length of the arc of the circle which forms the circumference of the cone's base

b the radius of the cone's base

c the slant height of the cone

d the total surface area of the cone, including the base.

12 Find the radius of each solid if it has a surface area of 6000 mm^2. Give your answer correct to 3 significant figures. R

a a sphere **b** a closed hemisphere **c** an open hemisphere

13 A cone has a surface area of 2000 cm^2. If the area of its base is 150 cm^2, find: R

a the radius of its base **b** its slant height **c** its perpendicular height

Mental skills 3: Maths without calculators ANSWERS ON P. 584

Quiz
Mental skills 3

Time differences

1 Study each example.

a What is the time difference between 11:40 a.m. and 6:15 p.m.?

From 11:40 a.m. to 5:40 p.m. = 6 hours

Count: '11:40, 12:40, 1:40, 2:40, 3:40, 4:40, 5:40'

From 5:40 a.m. to 6:00 p.m. = 20 min

From 6:00 p.m. to 6:15 p.m. = 15 min

6 hours + 20 min + 15 min = 6 hours 35 min

OR:



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b What is the time difference between 20:30 and 01:20?

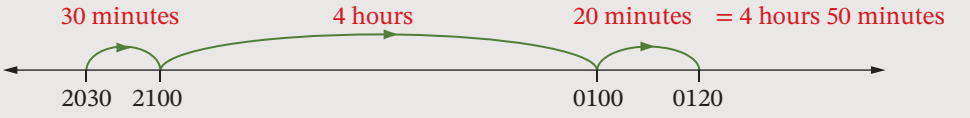
From 20:30 to 00:30 = 4 hours (24 − 20 = 4)

From 00:30 to 01:00 = 30 min

From 01:00 to 01:20 = 20 min

4 hours + 30 minutes + 20 minutes = 4 hours 50 minutes

OR:



2 Now find the time difference between:

a 11:10 a.m. and 7:40 p.m.

b 6:20 p.m. and 12:00 midnight

c 4:45 p.m. and 8:10 p.m.

d 2:35 a.m. and 10:50 a.m.

e 1:05 p.m. and 12:30 a.m.

f 9:35 a.m. and 11:15 a.m.

g 04:25 and 09:35

h 14:40 and 20:25

i 7:55 a.m. and 3:50 p.m.

j 2:40 p.m. and 10:20 p.m.

3.08 Surface areas of composite solids

Video
The power of the Sun

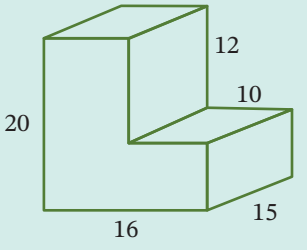
Worksheet
A page of prisms and cylinders

When calculating surface areas of composite solids, remember **not** to include the areas common to both solids.

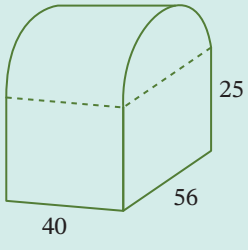
Example 16

Find, correct to one decimal place, the surface area of each solid. All measurements are in centimetres.

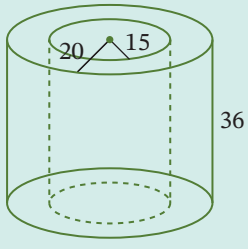
a



b

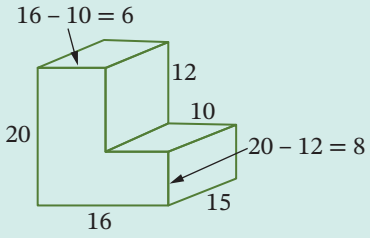


c



Solution

a This prism has 8 faces: 2 identical L-shapes (front and back) and 6 different rectangles.



Area of L-shape $= 16 \times 20 - 10 \times 12$

$= 200 \text{ cm}^2$

Surface area $= (2 \times 200) + (6 \times 15) + (12 \times 15) + (10 \times 15) + (8 \times 15) + (16 \times 15) + (20 \times 15)$

$= 1480 \text{ cm}^2$

Note that the 6 rectangles can also be thought of as one long rectangle of width 15 cm.

Length of long rectangle = perimeter of L = 72

Surface area $= (2 \times 200) + (72 \times 15)$

$= 1480 \text{ cm}^2$

b The solid is made up of a half-cylinder (3 faces) and a rectangular prism (5 faces).

Surface area of half-cylinder
= 2 semi-circular ends + curved surface area
$= 2 \times \frac{1}{2} \times \pi \times 28^2 + \frac{1}{2} \times 2 \times \pi \times 28 \times 40$
$= 5981.5924... \text{ cm}^2$

Radius of semi-circle $= \frac{1}{2} \times 56$
$= 28$

Do not round this partial answer, or the final answer will be inaccurate.

Surface area of rectangular prism $= (2 \times 40 \times 25) + (2 \times 56 \times 25) + (40 \times 56)$

$= 7040 \text{ cm}^2$

Total surface area $= 5981.5924... + 7040$

$= 13\,021.5924...$

$\approx 13\,021.6 \text{ cm}^2$

c The hollow cylinder is made up of 2 annulus (ring) faces, an outside curved surface area and an inside curved surface area.

Surface area of annulus faces $= 2 \times (\pi \times 20^2 - \pi \times 15^2)$

$2 \times$ area between 2 circles

$= 1099.5574... \text{ cm}^2$

Outside curved surface area $= 2 \times \pi \times 20 \times 36$

$= 4523.8334... \text{ cm}^2$

Inside curved surface area $= 2 \times \pi \times 15 \times 36$

$= 3392.9200... \text{ cm}^2$

Total surface area $= 1099.5574... + 4523.8934... + 3392.9200...$

$= 9016.3709...$

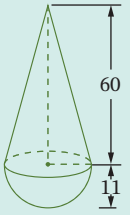
$= 9016.4 \text{ cm}^2$

PATH

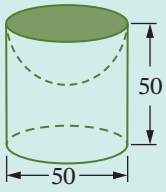
Example 17

Find, correct to the nearest square centimetre, the surface area of each solid.
All measurements are in centimetres.

a



b



Solution

a Surface area = curved surface area of cone + curved surface area of hemisphere

$$= \pi rl + \frac{1}{2} \times 4\pi r^2$$

$r = 11$, $h = 60$ and $l = ?$

$l^2 = 11^2 + 60^2$

$= 3721$

$l = \sqrt{3721}$

$= 61$ cm

Surface area $= \pi \times 11 \times 61 + \frac{1}{2} \times 4 \times \pi \times 11^2$

$= 2868.2740 \ldots$

≈ 2868 cm^2

b Surface area = curved surface of cylinder + circular base + curved surface of hemisphere

$= 2\pi rh + \pi r^2 + \frac{1}{2} \times 4\pi r^2$

$= 2\pi rh + 3\pi r^2$ $\qquad \pi r^2 + 2\pi r^2 = 3\pi r^2$

$= 2 \times \pi \times 25 \times 50 + 3 \times \pi \times 25^2$ $\qquad r = \frac{1}{2} \times 50 = 25$

$= 13\,744.4678 \ldots$

$\approx 13\,744$ cm^2

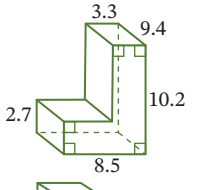
EXERCISE 3.08 ANSWERS ON P. 584

Surface areas of composite solids U F PS R

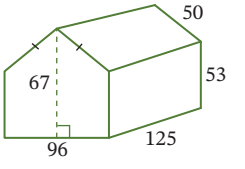
EXAMPLE 16

1 Find the surface area of each solid. All measurements are in centimetres. PS R

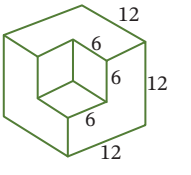
a



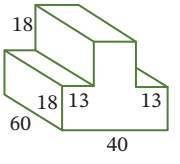
b



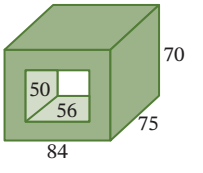
c



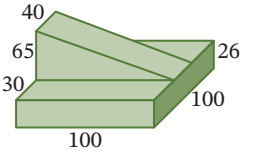
d



e



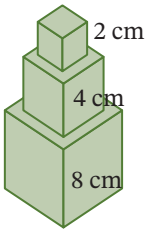
f



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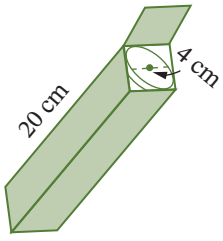
2 3 cubes of length 2 cm, 4 cm and 8 cm are glued on top of each other. Calculate the surface area of the new solid. PS R



3 Circular cracker biscuits of diameter 4 cm are packed in a cardboard box of length 20 cm.

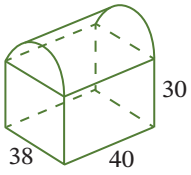
a Calculate the surface area of the box.

b How much cardboard would be saved if the biscuits were packed into a cylindrical box?

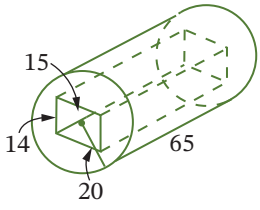


4 Find, correct to one decimal place, the surface area of each solid. All measurements are in centimetres. PS R

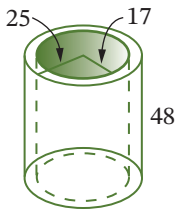
a



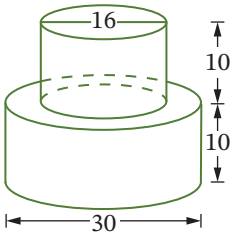
b



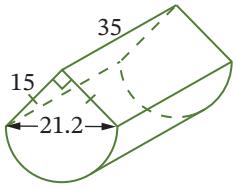
c



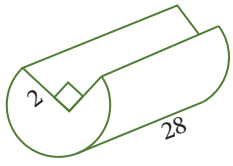
d



e



f

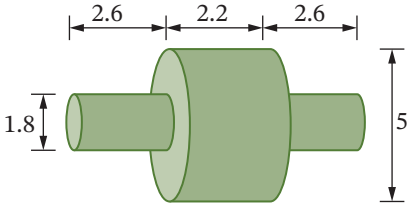


5 Calculate the surface area of the solid, correct to one decimal place. All measurements are in centimetres. R

Select the correct answer **A**, **B**, **C** or **D**.

A 86.0 cm^2 **B** 103.2 cm^2

C 108.3 cm^2 **D** 113.4 cm^2



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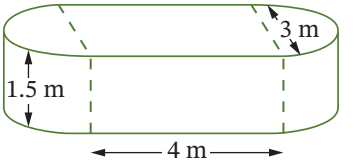
6 A wedding cake with 3 tiers rests on a table. Each tier is 6 cm high. The layers have radii of 20 cm, 15 cm and 10 cm respectively. Find the total visible surface area, correct to the nearest cm². PS R



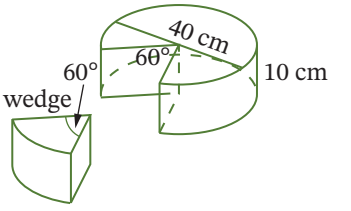
Shutterstock.com/Brian Barney

7 **a** Find, correct to 2 decimal places, the total external area of the wall of this above-ground swimming pool.

b Calculate the area of the water surface, correct to the nearest m². PS R



8 A wedge of cheese is cut from a cylindrical block of height 10 cm and diameter 40 cm. Find the total surface area of the wedge, correct to 2 decimal places. PS R



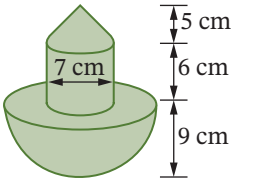
PATH

EXAMPLE 17

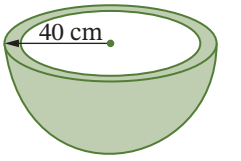
9 Calculate the surface area of the solid. Select **A**, **B**, **C** or **D**. PS R

A 923.98 cm² **B** 962.46 cm²

C 736.62 cm² **D** 669.51 cm²



10 Calculate the total surface of a hemispherical bird bath that is 1 cm thick and has an outer radius of 40 cm. PS R

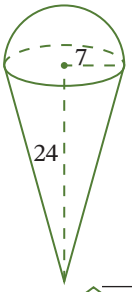


Foundation Standard Complex

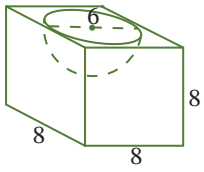
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11 Find, correct to one decimal place, the surface area of each solid. All measurements are in centimetres. PS R

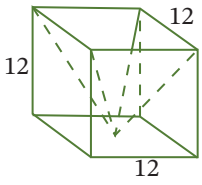
a



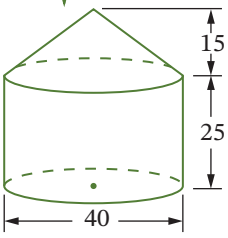
b



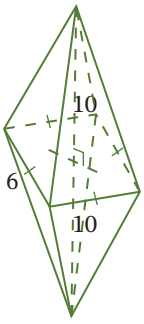
c



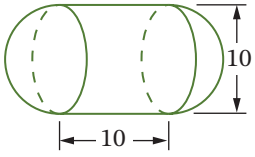
d



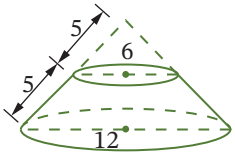
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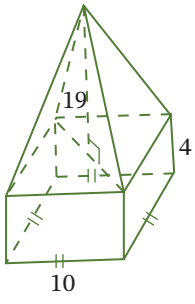
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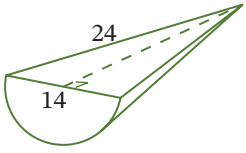
g



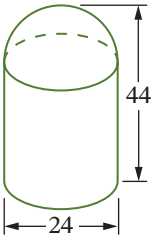
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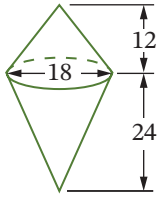
i



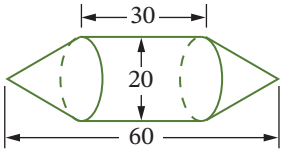
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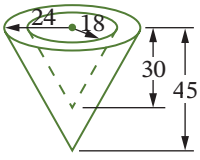
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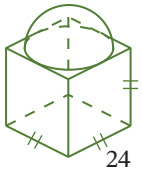
l



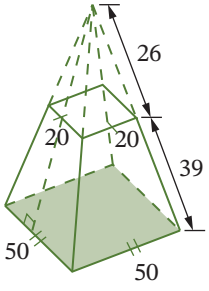
m



n



o



Foundation | Standard | Complex

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3.09 Volumes of prisms and cylinders

Worksheets
A page of prisms and cylinders
Volumes of solids
Back-to-front problems
Volume and capacity
Biggest volume

Technology
Volume of a box

Spreadsheet
Volume of a box

Puzzle
Formula matching game

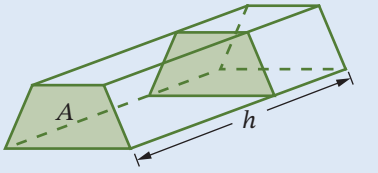
Videos
Capacity of a cylinder
Volumes of prisms and cylinders
Bees and their hives
The incredible strength of ants
The Menger sponge

Volume of a prism

$V = Ah$

where A = area of base

h = perpendicular height

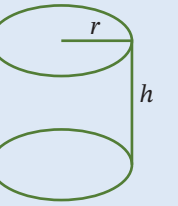


Volume of a cylinder

$V = \pi r^2 h$

where r = radius of circular base

h = perpendicular height



Volume and capacity

1 cm^3 contains 1 mL

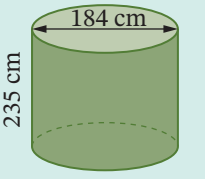
1 m^3 contains 1000 L or 1 kL

Example 18

For this cylinder, calculate:

a its volume correct to the nearest cm^3

b its capacity in kL, correct to 1 decimal place.



Solution

a Radius $= \frac{1}{2} \times 184$
$= 92$ cm
$V = \pi \times 92^2 \times 235$
$= 6\,248\,753.451...$
$\approx 6\,248\,753$ cm^3

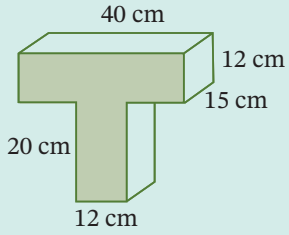
b Capacity $= 6\,248\,753$ mL
$= 6\,248\,753 \div 1000 \div 1000$ kL
$= 6.248\,753$ kL
≈ 6.2 kL

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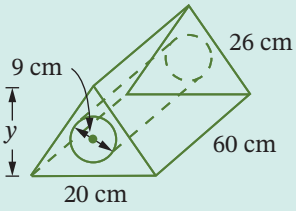
Example 19

Find, correct to the nearest whole number, the volume of each solid.

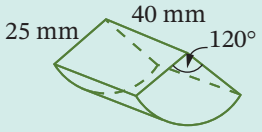
a



b



c



Solution

a $A = 40 \times 12 + 20 \times 12$

$= 720 \text{ cm}^2$

$V = Ah$

$= 720 \times 15$

$= 10\,800 \text{ cm}^3$

b Cross-section is a triangle minus a circle.

$26^2 = y^2 + 10^2$

$y^2 = 26^2 - 10^2$

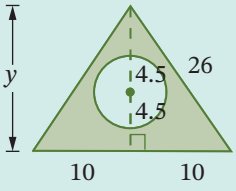
$= 576$

$y = 24 \text{ cm}$

Radius of circle $= \frac{1}{2} \times 9 = 4.5$

$A = \frac{1}{2} \times 20 \times 24 - \pi \times 4.5^2$

$= 176.3827... \text{ cm}^2$



$V = Ah$

$= 176.3827... \times 60$

$= 10\,582.9649...$

$\approx 10\,583 \text{ cm}^3$

c $A = \frac{120}{360} \times \pi \times 25^2$

$= 654.498... \text{ mm}^2$

$V = Ah$

$= 654.498... \times 40$

$= 26\,179.938...$

$\approx 26\,180 \text{ mm}^2$

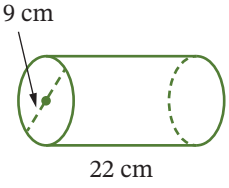
EXERCISE 3.09 ANSWERS ON P. 584

Volumes of prisms and cylinders UFR

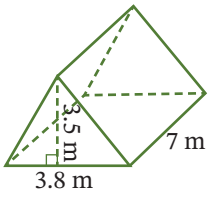
1 Calculate, correct to one decimal place, the volume of each solid.

EXAMPLE 18

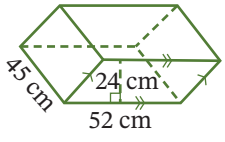
a



b

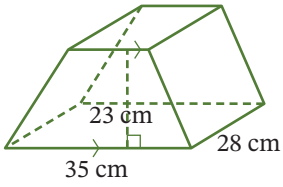


c

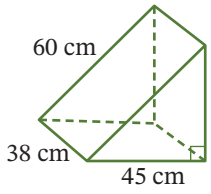


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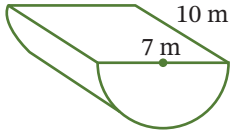
d



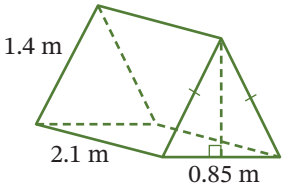
e



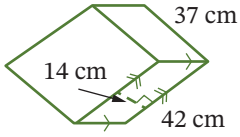
f



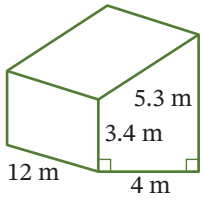
g



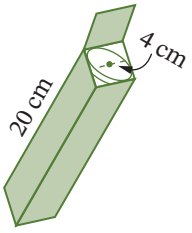
h



i

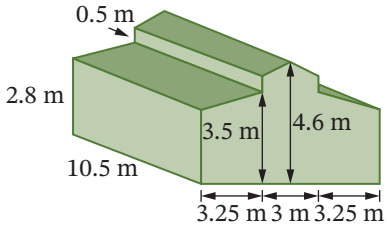


2 Rice crackers of diameter 4 cm are packed in a cardboard box of height 20 cm. Calculate, correct to one decimal place:

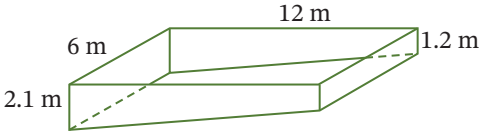


- **a** the volume of the crackers in the box
- **b** the volume of the box
- **c** the percentage of the box that is empty space.

3 Calculate, correct to one decimal place, the volume of the shed. R



4 This swimming pool is 12 m long and 6 m wide. The depth of the pool ranges from 1.2 m to 2.1 m.

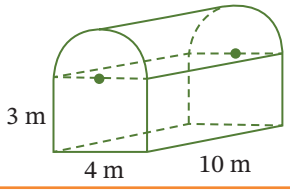


- **a** Calculate the capacity of this pool in litres.
- **b** If the pool is filled so that the water is 8 cm from the top of the pool, calculate the amount of water in the pool (to the nearest litre).

5 An Olympic sized swimming pool is 50 m long, 25 m wide and 2 m deep. What is the capacity of an Olympic pool in litres?

6 **a** Find, correct to 2 decimal places, the volume of this greenhouse.

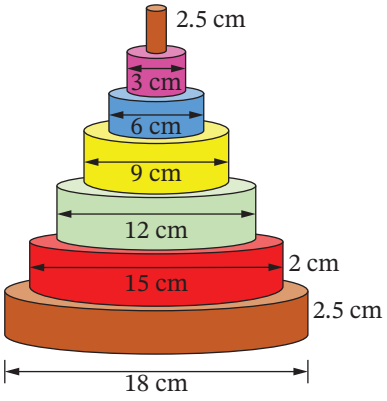
b If this greenhouse costs 0.5c per m^3 per hour to heat, how much is this per day (correct to the nearest cent)?



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7 A baby's toy involves placing 5 coloured cylinders on a wooden stick (diameter 1 cm) on a base (diameter 18 cm). The cylinders are of varying diameters with a height of 2 cm. Find the volume of the toy, including the base and peg, correct to one decimal place. R

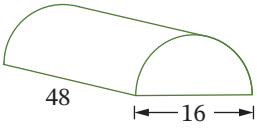


3.09

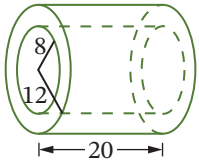
EXAMPLE 19

8 Find, correct to 2 decimal places where appropriate, the volume of each solid. All lengths shown are in centimetres. R

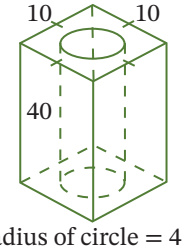
a



b

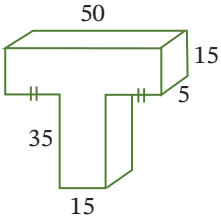


c

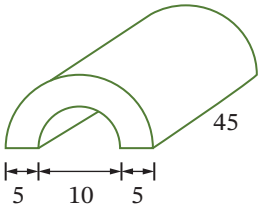


radius of circle = 4 cm

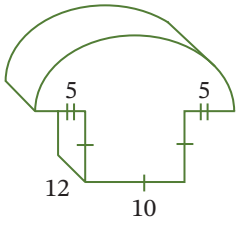
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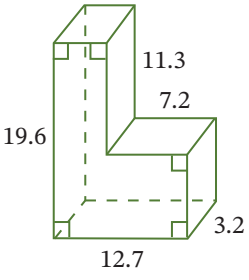
e



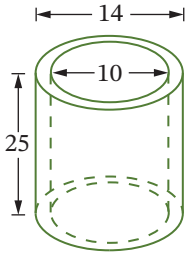
f



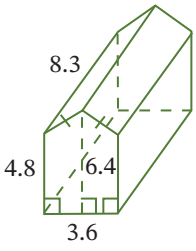
g



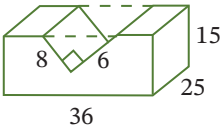
h



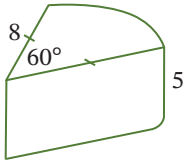
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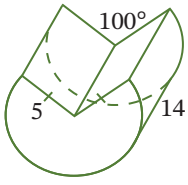
j



k



l



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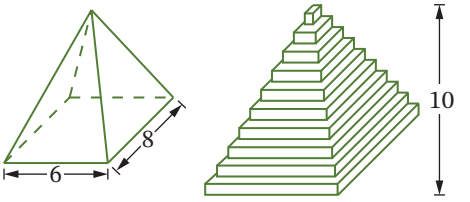
PATH

Technology
Approximating the volume of a cone

Technology

Approximating the volume of a pyramid

In this activity, we will use a spreadsheet to approximate the volume of a rectangular pyramid by slicing it into tiny layers of rectangular prisms of equal thickness.



Let $L = 6$ be the length of the prism, $W = 8$ be the width and $H = 10$ be the height.

The thickness, T, of each layer is given by the formula $T = \dfrac{H}{\text{number of layers}}$.

Starting at the bottom, the length and width of each layer are decreased by the amounts $\dfrac{L}{\text{number of layers}}$ and $\dfrac{W}{\text{number of layers}}$ with each step.

1 Set up your spreadsheet as shown.

	A	B	C	D	E	F
1						
2			Number of layers =			
3	H	L	W	Thickness of layers	Volume of layer	Sum of volumes
4	10	6	8	=$A$4/$D$2	=B4*C4*D4	=E4
5		=B4-$B$4/$D$2	=$C4-$C$4/$D$2			=E5+F4
6						
7						

2 Let the number of layers be 10. Enter 10 in cell D2.

3 Copy each formula down to row 13.

4 Explain the results in cells E13 and F13.

5 How accurate was your result in F13? Explain.

6 Print out your spreadsheet and paste it into your book.

7 Enter 40 (layers) in cell D2 and copy each formula down to row 43.

8 In 1–2 sentences compare your result in F43 with the previous result in F13 from question 4.

9 Enter each value in cell D2 and copy down the formulas as requested.

a 100 (copy down to row 103)

b 200 (copy down to row 203)

c 400 (copy down to row 403)

10 Use the formula $V = \frac{1}{3}Ah$ to calculate the exact volume of the pyramid.

11 Write a brief report about your results in questions **9** and **10**.

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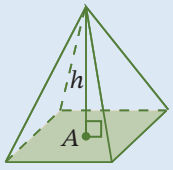
Volumes of pyramids, cones and spheres 3.10

Volume of a pyramid

$$V = \frac{1}{3}Ah$$

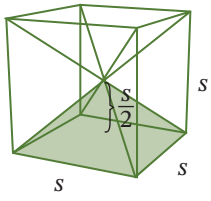
where A = area of the base

h = perpendicular height.



The volume of a pyramid is $\frac{1}{3}$ of the volume of a prism with the same base and height.

This fact is difficult to prove mathematically, but here's a diagram that might help. This is a square pyramid with dimensions s, s and $\frac{s}{2}$, with an identical pyramid balancing on top of it upside-down. Now look at the big cube that surrounds these 2 square pyramids.

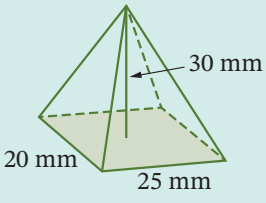


- Can you see how many such pyramids fit into this big cube exactly?
- What is the volume of the big cube?
- So what is the volume of one pyramid?
- Is this the same as $V = \frac{1}{3}Ah$?

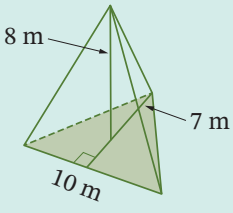
Example 20

Find the volume of each pyramid.

a



b



Solution

a

$A = 20 \times 25$

$= 500$

$V = \frac{1}{3}Ah$

$= \frac{1}{3} \times 500 \times 30$

$= 5000 \text{ mm}^3$

b

$A = \frac{1}{2} \times 10 \times 7$

$= 35$

$V = \frac{1}{3}Ah$

$= \frac{1}{3} \times 35 \times 8$

$= 99\frac{1}{3} \text{ m}^3$

PATH

3.10

Video
The Pacific Flyer

Worksheet
Back-to-front problems (Advanced)

Video
Volume of a pyramid

PATH

Example 21

Find the volume of a square pyramid with base length 22 mm and slant height 61 mm.

Solution

First find h, the perpendicular height of the pyramid.

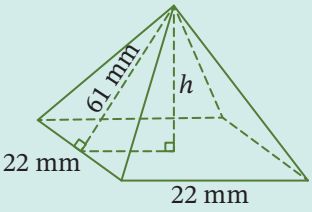
$h^2 = 61^2 - 11^2 = 3600$

$h = 60$ mm

$A = 22^2 = 484$

$V = \frac{1}{3} \times 484 \times 60$

$= 9680$ mm^3



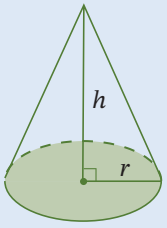
Volume of a cone

Video
Surface area and volume of a cone

$$V = \frac{1}{3}\pi r^2 h$$

where r = radius of the base

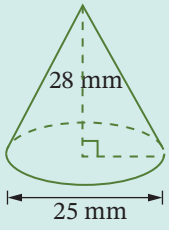
h = perpendicular height.



A cone is like a 'circular pyramid' so $V = \frac{1}{3}Ah = \frac{1}{3} \times \pi r^2 \times h = \frac{1}{3}\pi r^2 h$

Example 22

Find, correct to the nearest cubic centimetre, the volume of this cone.



Solution

$V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \pi \times 12.5^2 \times 28 \quad r = \frac{1}{2} \times 25 = 12.5$

$= 4581.4892...$

≈ 4581 cm^3

PATH

3.10

Example 23

A cone has a slant edge of 50 mm and a base radius of 15 mm.
Find its volume correct to 2 significant figures.

Solution

$h^2 = 50^2 - 15^2$

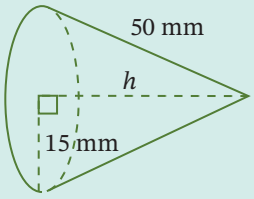
$= 2275$

$h = \sqrt{2275}$

$V = \frac{1}{3} \times \pi \times 15^2 \times \sqrt{2275}$

$= 11\,238.3314...$

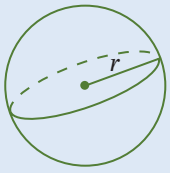
$\approx 11\,000 \text{ mm}^3$



Volume of a sphere

$$V = \frac{4}{3}\pi r^3$$

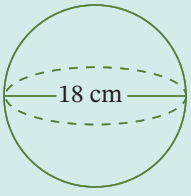
where r = radius of the sphere.



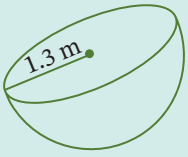
Example 24

Find, correct to 2 significant figures, the volume of each solid.

a



b



Solution

a $V = \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \pi \times 9^3$ $\quad r = \frac{1}{2} \times 18 = 9$

$= 3053.6280...$

$\approx 3100 \text{ cm}^3$

b $V = \frac{1}{2} \times \frac{4}{3}\pi r^3$

$= \frac{2}{3}\pi r^3$

$= \frac{2}{3} \times \pi \times 1.3^3$

$= 4.6013...$

$\approx 4.6 \text{ m}^3$

Video
Volume of a sphere

PATH

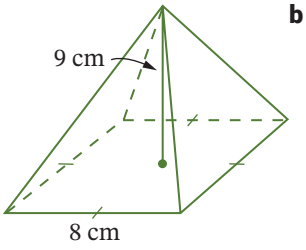
EXERCISE 3.10 ANSWERS ON P. 584

Volumes of pyramids, cones and spheres U F PS R C

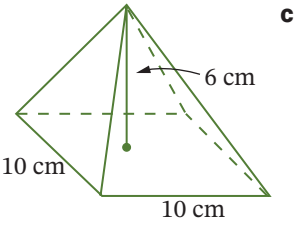
EXAMPLE 20

1 Find, correct to one decimal place, the volume of each pyramid.

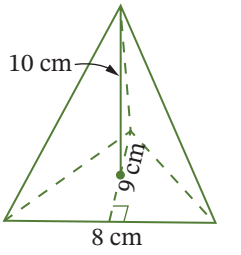
a



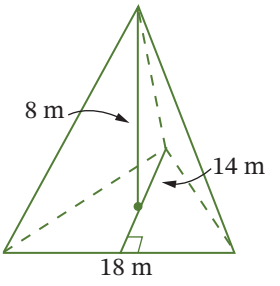
b



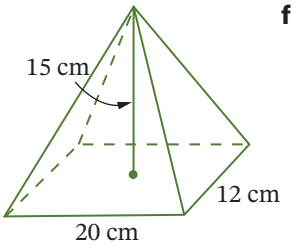
c



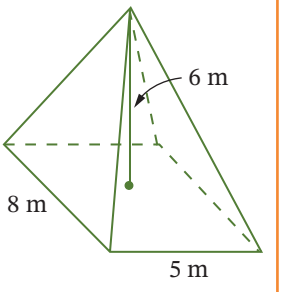
d



e



f

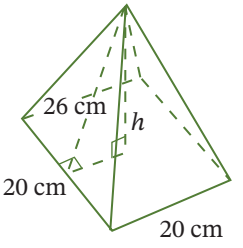


EXAMPLE 21

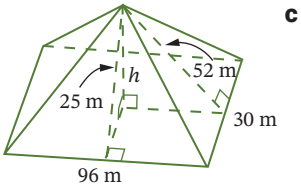
2 For each pyramid, find correct to one decimal place: R

i its perpendicular height **ii** its volume

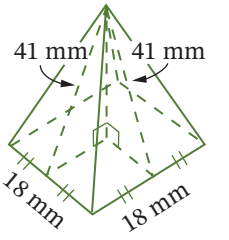
a



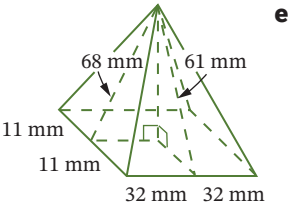
b



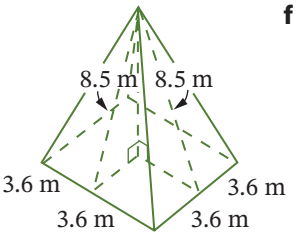
c



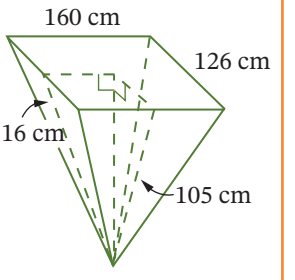
d



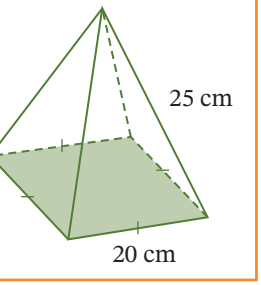
e



f



3 Calculate the volume of this square pyramid.
Select the closest answer **A**, **B**, **C** or **D**. R



A 3055.1 cm^3 **B** 2000 cm^3

C 3333.3 cm^3 **D** 2748.7 cm^3

Foundation Standard Complex

9780170479622

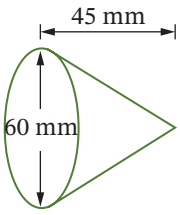
PATH

EXAMPLE 22

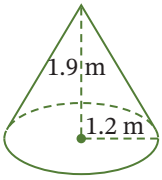
3.10

4 Find, correct to the nearest whole number, the volume of each cone.

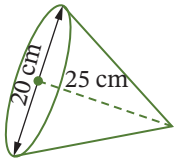
a



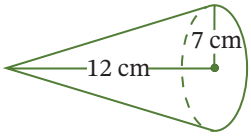
b



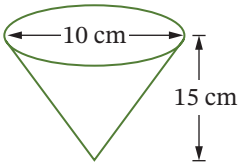
c



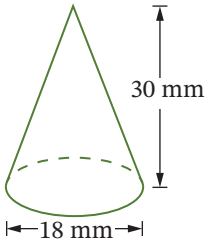
d



e



f

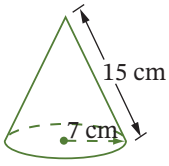


EXAMPLE 23

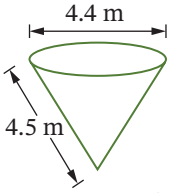
5 For each cone, find correct to one decimal place: R

i its perpendicular height **ii** its volume

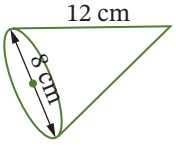
a



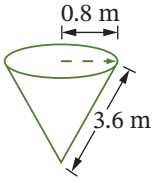
b



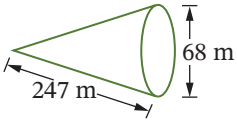
c



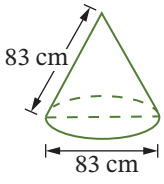
d



e



f



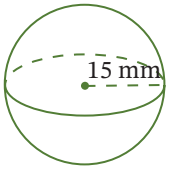
6 Find the volume of a cone with a base diameter of 5.4 cm and a slant height of 9.8 cm. Select **A**, **B**, **C** or **D**. R

A 75 cm^3 **B** 72 cm^3 **C** 288 cm^3 **D** 111.1 cm^3

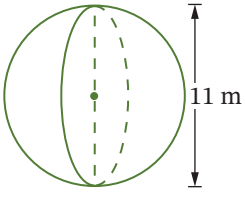
EXAMPLE 24

7 Find the volume of each sphere or hemisphere, correct to one decimal place.

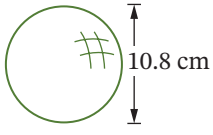
a



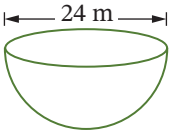
b



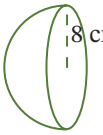
c



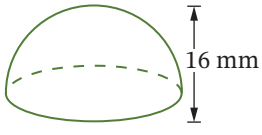
d



e



f

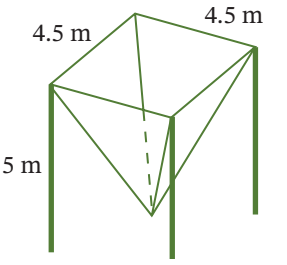


Foundation Standard Complex

PATH

8 The Earth has a radius of approximately 6378 km. Calculate its volume in scientific notation, correct to 3 significant figures. C

9 A grain hopper is in the shape of a square pyramid. PS R C



a Find the volume of grain it holds when full.

b If there are 750 kg of wheat per m^3, find the mass of grain in the hopper when it is filled to three-quarters of capacity. Give your answer correct to the nearest tonne.

10 A pyramid has a volume of 360 m^3 and a base area of 48 m^2. Calculate its perpendicular height. R

11 A square pyramid has a volume of 800 cm^3 and a perpendicular height of 12 cm. Calculate, correct to one decimal place, the length of its base. R

12 A cone has a volume of 600 m^3 and a base radius of 10 m. Calculate, correct to 1 decimal place, its perpendicular height. R

13 A cone has a volume of 160 cm^3 and a perpendicular height of 20 cm. Calculate, correct to one decimal place, its radius. R

14 The volume of a spherical light is 2664 cm^3. Calculate its diameter. Select **A**, **B**, **C** or **D**. R

A 17.2 cm **B** 8.6 cm **C** 17.0 cm **D** 8.5 cm

3.11 Volumes of composite solids

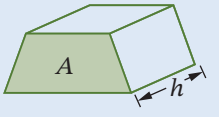
PATH

Presentation
Volumes of shapes

Volume formulas

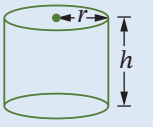
Prism

$V = Ah$



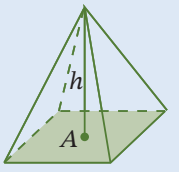
Cylinder

$V = \pi r^2 h$



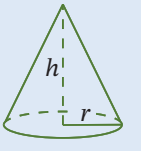
Pyramid

$V = \frac{1}{3}Ah$



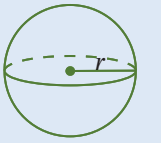
Cone

$V = \frac{1}{3}\pi r^2 h$



Sphere

$V = \frac{4}{3}\pi r^3$



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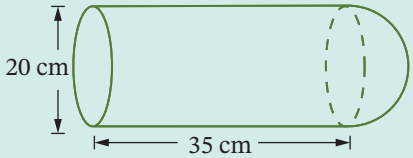
Note that the formulas for **surface area** involve **2 dimensions**, for example, r^2 or rh, while the formulas for **volume** involve **3 dimensions**, for example, r^2h or r^3 or Ah.

Example 25

For this packing capsule, find correct to the nearest whole number:

a its volume

b its capacity in litres



Solution

a Volume $= \pi r^2 h + \frac{1}{2} \times \frac{4}{3} \pi r^3$

$= \pi \times 10^2 \times 35 + \frac{2}{3} \times \pi \times 10^3$

$= 13\,089.9693...$

$\approx 13\,090 \text{ cm}^3$

b Capacity $\approx 13\,090$ mL

$= 13.09$ L

≈ 13 L

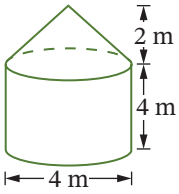
EXERCISE 3.11 ANSWERS ON P. 584

Volumes of composite solids U F PS R

1 This storage tank shown is completely filled with water.

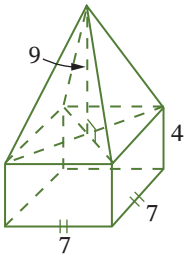
a Calculate to the nearest cubic metre the volume of the tank.

b Find the capacity of the tank, correct to the nearest kilolitre.

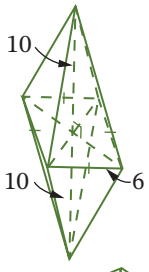


2 Find the volume of each solid. All measurements are centimetres. R

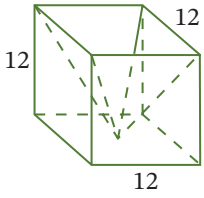
a



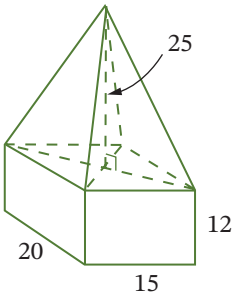
b



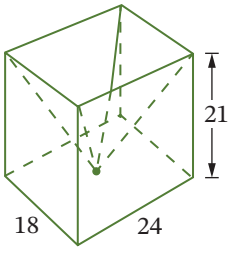
c



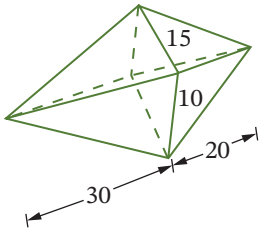
d



e



f



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PATH

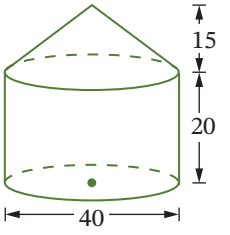
3 For each solid, find:

i the volume (to the nearest cm^3)

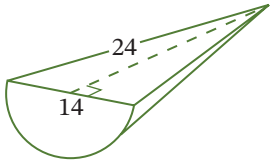
ii the capacity (in litres, correct to 3 decimal places).

All measurements are in centimetres.

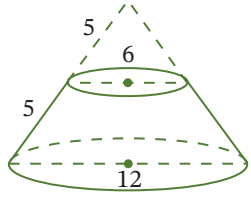
a



b

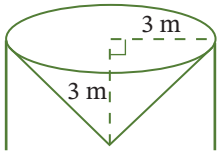


c

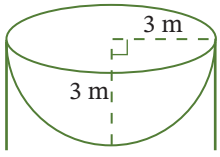


4 A conical tank (A) and a hemispherical tank (B) have measurements as shown. How much more does tank B hold compared to tank A? Answer correct to 2 decimal places.

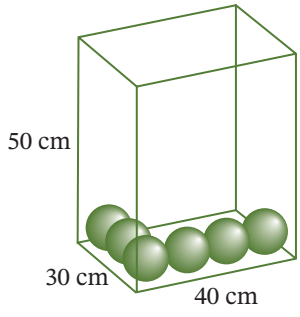
A



B



5 Spherical balls of diameter 10 cm are stacked inside a box as shown. PS R



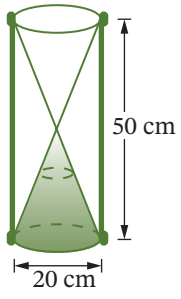
a How many balls will fit in the bottom layer?

b If the balls are stacked in the same manner as the bottom layer until the box is full, how many balls will fit in the box?

c Calculate the volume of the space occupied by the balls when the box is full.

d What percentage of the box is empty space? Give your answer to the nearest whole percentage.

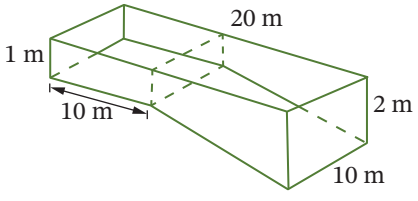
6 The sand in this hourglass takes up $\frac{3}{4}$ of the volume of the bottom cone. PS R



a Calculate, correct to the nearest cubic centimetre, the volume of sand in the hourglass.

b If the sand takes one hour to fall from the top cone to the bottom cone, at what rate is it falling? Give your answer in cm^3/s, correct to 2 significant figures.

7 **a** Calculate the volume of this swimming pool. PS R



b Calculate the capacity of the pool if it is filled to a depth 20 cm from the top.

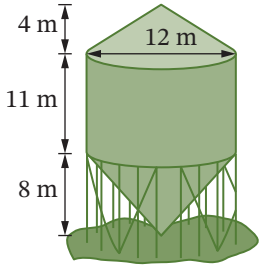
c If water costs \$1.98/kL, find the cost of filling the pool.

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8 **a** Find the capacity of the silo shown, correct to the nearest m^3. PS R

b How many tonnes of barley grain will the silo hold if the density of barley grain is 600 kg/m^3?



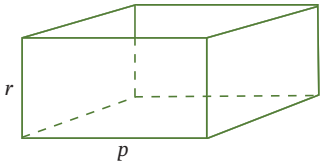
Power plus ANSWERS ON P. 584

1 The total surface area of a cube is 864 cm^2. Find its volume.

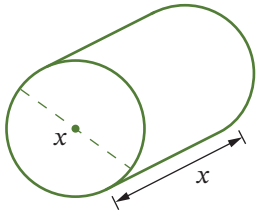
2 A cylinder has a volume of 3619.11 cm^3. Its height is 18 cm. Calculate to the nearest centimetre the radius of its base.

3 Find a formula for the surface area, SA, of:

a this open square prism of base length p and height r

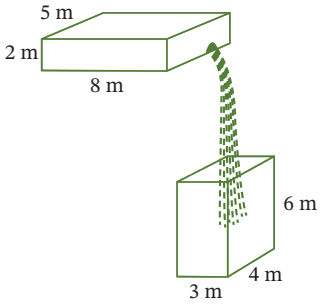


b this cylinder of diameter and height x

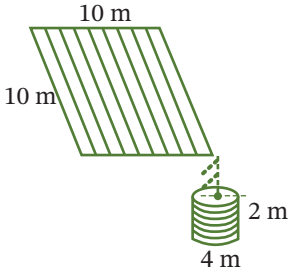


4 The surface area of the curved surface of a can is 27 143.4 mm^2. If its height is 120 mm, find the radius of the can.

5 Water flows from the top tank to the bottom tank at a constant rate. The level of water in the top tank falls at a rate of 15 cm/h. At what rate is the level of water rising in the bottom tank?



6 A 10 m flat square roof drains into a cylindrical rainwater tank with a diameter of 4 m. If 5 mm of rain falls on the roof, by how much (to the nearest millimetre) does the level of the water in the tank rise?



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CHAPTER 3 REVIEW

Quiz
Language of maths 3

Puzzle
Surface area and volume crossword

Language of maths

absolute error	apex	base	capacity
cone	cross-section	curved surface	cylinder
diameter	hemisphere	kilolitre	percentage error
perpendicular height	precision	prism	pyramid
radius	rounding error	sector	slant height
sphere	surface area	truncate	volume

1 What is the difference between rounding and truncating a number?

2 Which word means a 'slice' of a prism or cylinder?

3 Name 3 solids that have a **curved surface area**.

4 What is the formula for the curved surface area of a cone?

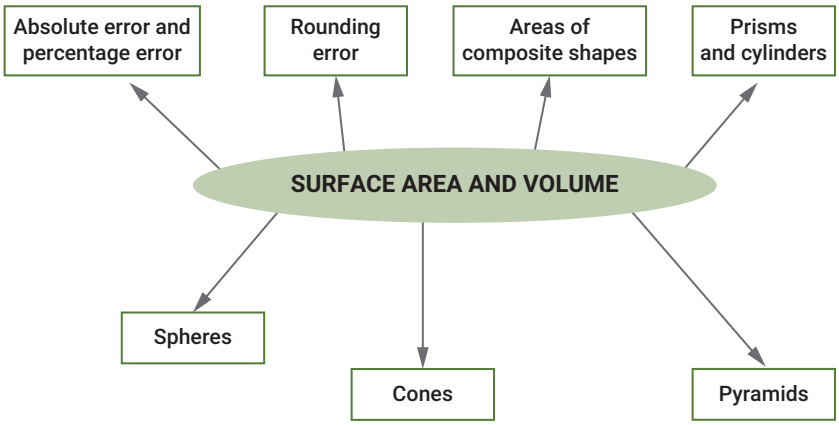
5 Explain the difference between the **perpendicular height** and the **slant height** of a pyramid.

6 What is the formula $V = \frac{1}{3}\pi r^2 h$ used for?

Topic summary

Worksheet
Mind map: Surface area and volume

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



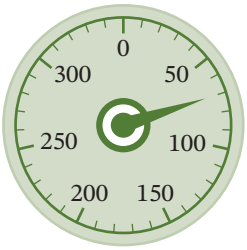
9780170479622

TEST YOURSELF 3

ANSWERS ON P. 585

Quiz
Test yourself 3

1 This weight scale shows a measurement of 70 g. Find the measurement's: 3.01



a absolute error

b percentage error (correct to one decimal place)

2 A plant stem is measured to be 16.0 cm correct to 1 decimal place. What is the percentage error in this measurement? 3.01

A 0.3125 % **B** 0.625 % **C** 3.125 % **D** 6.25 %

3 A triangle has a base length of 5.65 cm and a perpendicular height of 11.31 cm. 3.02

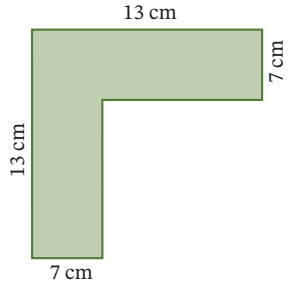
a Calculate its area.

b Round the triangle's base and height to one decimal place, use these values to calculate its area, then find correct to one decimal place the percentage error of the calculated area.

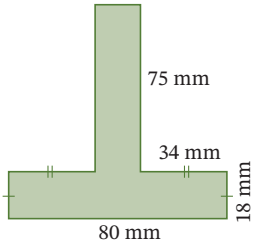
c Truncate the triangle's base and height to the nearest whole number, use these values to calculate its area, then find correct to one decimal place the percentage error of the area.

4 Find the area of each shape. Give your answers correct to one decimal place where necessary. 3.03

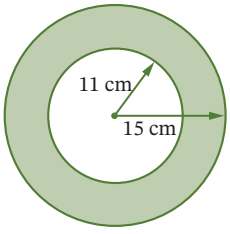
a



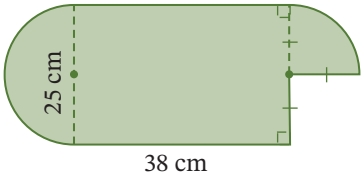
b



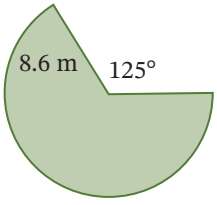
c



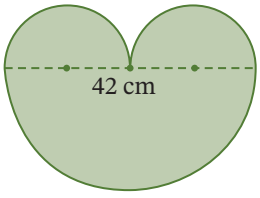
d



e



f

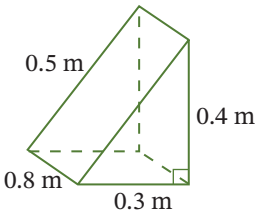


Foundation | Standard | Complex

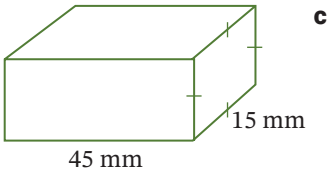
3.04

5 Find the surface area of each prism.

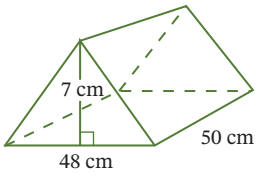
a



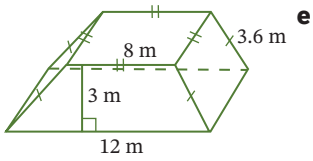
b



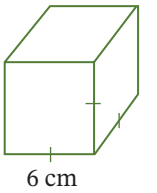
c



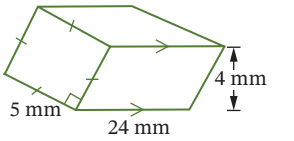
d



e



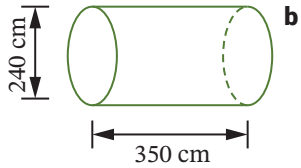
f



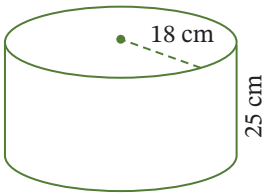
3.05

6 Calculate, correct to one decimal place, the surface area of each solid.

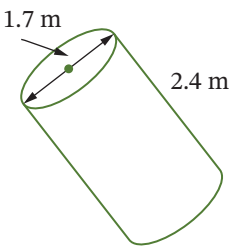
a



b

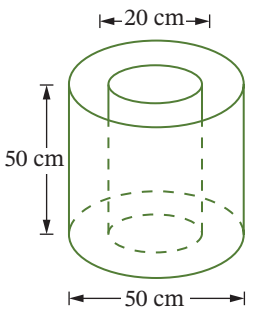


c

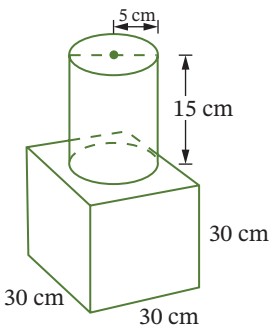


Cylinder, open at one end

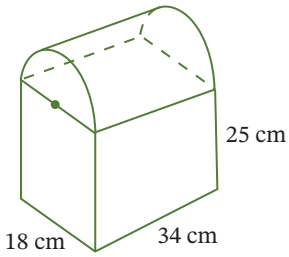
d



e



f



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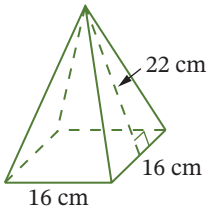
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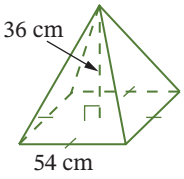
3.06

7 Find the surface area of each pyramid.

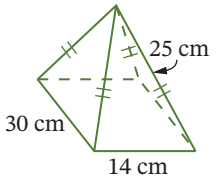
a



b



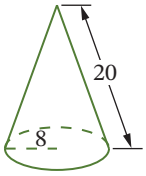
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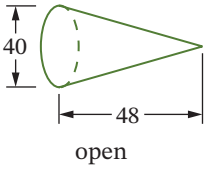
3.07

8 Find the surface area of each solid. All measurements are in metres.

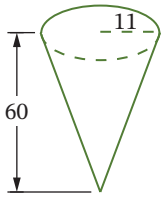
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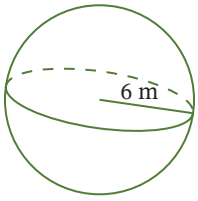
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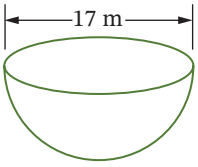
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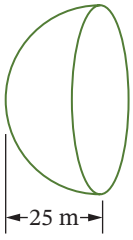
d



e



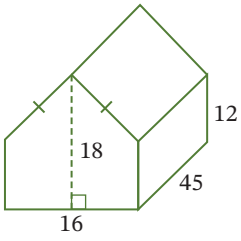
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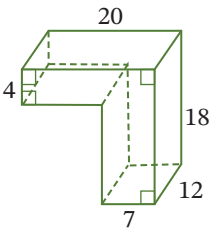
3.08

9 Find, correct to the nearest square centimetre, the surface area of each solid. All measurements are in centimetres.

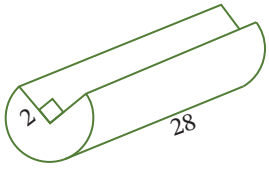
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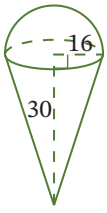
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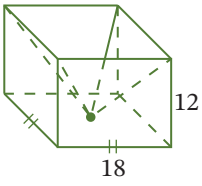
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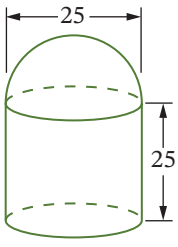
d



e



f



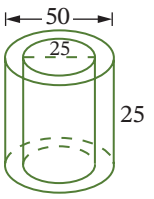
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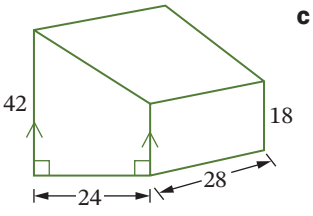
3.09

10 Calculate, correct to nearest cubic metre, the volume of each solid. All measurements are in metres.

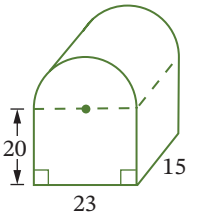
a



b



c

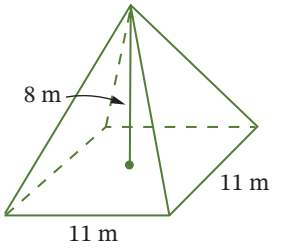


PATH

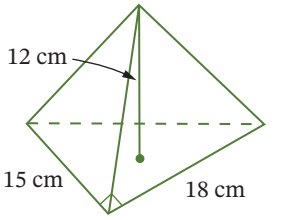
3.10

11 Find, correct to 2 decimal places (where necessary), the volume of each solid.

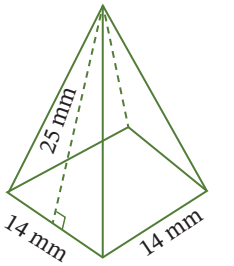
a



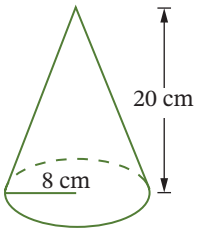
b



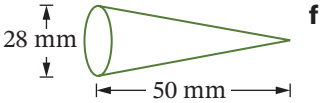
c



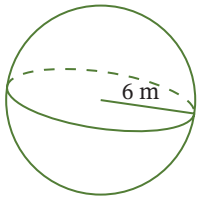
d



e



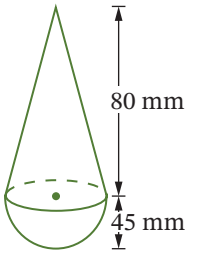
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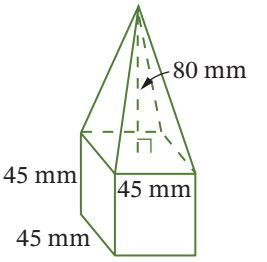
3.11

12 Find, correct to the nearest whole number, the volume of each solid.

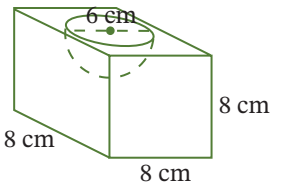
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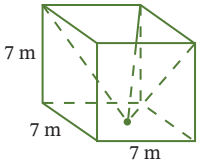
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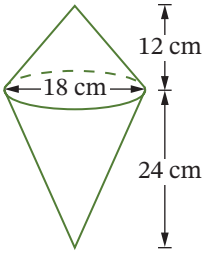
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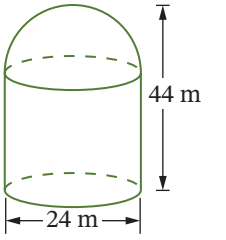
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